PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. Under the Panjab University Act VII of 1947—enacted by the Govt. of India)

SYLLABI

FOR

B.Sc./B.Sc. (Hons)/B.Sc. (Hons. with research)
Discipline Specific Core Papers

FIRST YEAR
(SEMESTER SYSTEM)
EXAMINATIONS, 2026-27

(SEMESTER : FIRST AND SECOND)

i.e

First Semester: ***November/December, 2026***
Second Semester: ***April/May, 2027***

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Subject combinations offered in the Faculty of Science under NEP

1 - Elective	**2 -** Elective	**3 -** Elective	**4 -** Elective	**5 -** Elective
Physics	Chemistry	Mathematics	Bio-informatics	Computer Science and Applications
Botany	Statistics	Zoology	Biotechnology	
			Industrial Chemistry	
			Agriculture	
			Fashion Designing	
			Bio-chemistry	
			Microbiology	
			Microbial Food Technology	

Note: Candidate can opt total three subjects choosing only one subject from each from the column No. 1 to 5.

Table-2

NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY B.SC, B.SC HONOURS AND B.SCHONOURS WITH RESEARCH

Semester	Subject- A (4 Cr)	Subject-B (4 Cr)	Subject- C (4 Cr)	MDC /IDC (3 Cr)	AEC (2 Cr)	VAC (2 Cr)	SEC (3 Cr)	Internship/ Community Work (4 Cr)	Pbi/ HCP (2Cr)	Total
I	MAJ A-1	MAJ B-1	MIN C-1	MDC-1	Eng-1	VAC-1	SEC-1		Pbi-1/ HCP-1	24
II	MAJ A-2	MAJ B-2	MIN C-2	MDC-2	Eng-2	VAC-2	SEC-2		Pbi-2/ HCP-2	24
The student exiting the programme after one year, securing 48 credits, will be awarded UG Certificate provided he/she secures 4 credits from summerinternship/community work.										
The candidate will have to declare one of the three subjects as Minor in the 3rd semester. The other 02 subjects will remain his/her Major subjects										
III	MAJ A-3	MAJ B-3	MIN C-3 MIN C-4	MDC-3	Eng / MIL		SEC-3			24
IV	MAJ A-4 MAJ A-5	MAJ B-4 MAJ B-5	MIN C-5		Eng/ MIL	VAC-3				24
The student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secures 4 credits from summer internship										
V	MAJ A-6 MAJ A-7	MAJ B-6 MAJ B-7	MIN C-6					Internship/ Community Work		24
VI	MAJ A-8 MAJ A-9	MAJ B-8 MAJ B-9	MIN C-7 MIN C-8							24
The student successfully securing 144 credits will be awarded BSc degree										
	36	36	32	09	08	06	09	04	04	144
VII	MAJ A-10 MAJ A-11	MAJ B-10 MAJ B-11	MIN C-9 MIN C-10							24
VIII	MAJ A-12 OR RM MAJ A – 13 MAJ A – 14 Or Dissertation of 08 credits	MAJ B-12 OR RM MAJ B – 13 MAJ B – 14 Or Dissertation of 08 credits								24
	56	56	40	09	08	06	09	04	04	192

Notes:-

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at end of the second semester.
3. The student choosing Multi-Disciplinary stream will continue with the three subjects and complete his/her three year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3 year degree) shall be eligible for 4 YEAR DEGREE B.SC HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain at least 12 credits in that subject(s).

6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two year securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

Foreword

This document provides the syllabi for the first two semesters of various science subjects in the category of Discipline Specific Courses (DSC) for the first year students enrolled in four year degree programme in Science stream (except B.Sc. Biochemistery, B.Sc. Bioinformatics, B.Sc. Biotechnology, B.Sc Microbial and Food Technology and BCA) under NEP. The Syllabi of other categories i.e. MDC, AEC, VAC, SEC, Copulsory Punjabi and HCP have been given separately.

The University has approved two distinct templates leading to two types of degrees: Subject-Specific and Discipline-Specific. This document is alligned with the Template-2 which is Multidisciplinary Stream.

Agriculture

NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY B.A./ B.Sc., B.A./B.Sc. Honours and B.A./B.Sc. Honours with Research in Agriculture

Sem	Level	Course Code	Course Name	Credits (Th+Pr+Tut)	Total Credits
I	L-100	AGR-DSC-1-Maj/Min-101	Elements of Agronomy and Agriculture	3+1=4 (x3)	24
		AGR-MDC-1-101	Agricultural Botany & Crop Physiology	2+1=3	
		AGR-VAC-1-101	Environment Education	2	
		AGR-SEC-1-101	Nursery Production	2+1=3	
		AEC-1-101	Eng-1	2	
		PUN-COMPL-1-101/HCP-COMPL-1-101	Pbi-1/HCP-1	2	
II	L-100	AGR-DSC-2-Maj/Min-201	Basics of Agricultural Economics	3+1=4 (x3)	24
		AGR-MDC-2-201	Agricultural Microbiology	2+1=3	
		AGR-VAC-2-201	Sustainable Vegetable Crops for kitchen gardening	1+1	
		AGR-SEC-2-201	Biofertilizers	2+1=3	
		AEC-2-201	Eng-2	2	
		PUN-COMPL-2-201/HCP-COMPL-2-201	Pbi-2/HCP-2	2	
The candidate has to select Major and Minor out of 3 subjects studied in the I and II Semester at the end of the Semester II					
The student exiting the programme after one year, securing 48 credits, will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work.					

Notes:

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre

5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

B.A./B.Sc.(General) Agriculture (Major/Minor)

Course Title: ELEMENTS OF AGRONOMY AND AGRICULTURE **Course Code: (AGR-DSC-1 Maj/Min-101)**
Semester: First

Learning Outcome: To give student elementary information on basic agronomic principles and practices.

Max. Marks: 100 **Course Credits** = 3+1=4

1. Theory: 75	Written Exam: 67	Internal Assessment: 8
2. Practical: 25	Written Exam: 22	Internal Assessment: 3

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
2. Practical- One of two hours duration

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 10 short answer type questions of one and a half mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teachers can use lecture, lecture-cum-demonstration, project, assignments, problem solving, inductive and deductive methods and can use L.C.D., L.E.D., and multimedia projector in addition to black/white board.

UNIT – I 12 Hours

Evolution of agriculture, ancient agriculture, history of Indian Agriculture, Principals of agronomy, classification of crops.
Crop rotation: principles, limitation, advantages; rotational intensity, cropping scheme, cropping intensity.
Cropping system: intercropping, mixed cropping, multiple cropping and relay cropping.

UNIT – II 11 Hours

Agronomy as a science and its relationships with other sciences.
Germination, maturity, harvesting and storage of crop plants.
Physical &biological properties of soil, soil organic matter.
Tillage principles, tillage practices for different soil types and crops.

UNIT – III 11 Hours

Seed bed preparation.
Characteristics of good seed beds.
Methods of sowing and their suitability under different conditions.
Seedling practices in relation to types of seeds, time of sowing and soil moisture.

UNIT – IV **11 Hours**

Weed characteristics, dissemination and competition for growth factors and problems caused by them.
Common weeds, losses and benefits, crop weed associations
Common methods of weed control.
Green manuring – advantages, limitations, green manure crops.

GUIDELINES FOR PRACTICAL EXAMINATION

Time- 3 Hrs **Max Marks: 25**
Practical: 22 **Internal Assessment: 03**

1. Land measurement
2. Identification and practices in the use of farm hand tools, bullock and tractor driven implements.
3. Identification of crop seeds, crop plants and important weeds.
4. Knowledge of various fertilizers – collection, weeding, hoeing in various crops.
5. Collection of weeds
6. Use of spray pumps.
7. Calculation of herbicide doses for different crops.

Suggested Readings:-

1. Reddy S.R., *Principles of Agronomy.* Kalyani Publishers, Ludhiana, 2018.
2. Handbook of Agriculture, I.C.A.R. Publications, New Delhi, 2008.
3. *Weeds of North India* I.C.A.R. Publications, New Delhi, 2008.
4. *Package of Practices for Rabi and Kharif crops,* P.A.U. Publications Ludhiana, Corresponding year.

B.A./B.Sc.(General) Agriculture Multi-Disciplinary

Course Title: AGRICULTURAL BOTANY AND CROP PHYSIOLOGY **Course Code: (AGR-MDC-1-101)**
Semester: First

Learning Outcome: To learn ecological concepts and theories of evolutions, growth and development of crops.

Max. Marks: 75 **Course Credits** = 2+1=3

1.	Theory: 50	Written Exam: 45	Internal assessment: 5
2.	Practical: 25	Written Exam: 22	Internal assessment: 3

Periods per week: 1. Theory- Three of 45 minutes duration (Two Hours)
2. Practical- One of two hours duration (Two Hours)

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 9 short answer type questions of one mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 9 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teachers can use lecture, lecture-cum-demonstration, project, assignments, problem solving, inductive and deductive methods and can use L.C.D., L.E.D., and multimedia projector in addition to black/white board.

UNIT – I **8 Hours**

Plant ecology, nature and scope of ecology, concept of environment.
Ecological factors and their influence on distribution of vegetation.
Main ecological plant communities and their adaptations.
Concept of ecosystem and biosphere.

UNIT – II **7 Hours**

Sources of pollution, environmental conservation and health.
Concept of center of origin of crop plants.
Domestication of crops and evaluation of germplasm.
Weed ecology and weed competitions.

UNIT – III **8 Hours**

Introductory concept of theories of organic evolution.
Evolution of important group of plants.
Variation, their types and causes, adaptations.
Growth and development of crops: Photosynthesis, respiration, net photosynthesis, translocation of assimilates.

UNIT – IV **7 Hours**

Growth of plants: development stages, physiological growth components, factors influencing growth and development.
Plant growth substances, growth expressions.
Transpiration, absorption of food material by plants.
Water absorption and translocation.

GUIDELINES FOR PRACTICAL EXAMINATION

Time: 3 hrs **Max Marks: 25**
Practical: 22 **Internal Assessment: 03**

1. Growth analysis experiments demonstrating various aspects of respiration, photosynthesis, and ascent of sap.
2. Water culture techniques/ deficiency and toxicity of major and minor elements.

Suggested Readings:

1. Devlin, R.M., *Plant Physiology*, Prindle Weber &SvhmidtPublisher, New York, 1983.
2. Kochhar, P.L., *Plant Physiology*, Trumen Publishers Jalandhar, 2010.
3. Bhatia K.N., and Widge, R., Foundation *of Botany*, Truman Publishers, Jalandhar, 2010.

B.A./B.Sc.(General) Agriculture Skill

Course Title: NURSERY PRODUCTION **Course Code: (AGR-SEC-1-101)**
Semester: First

Learning Outcomes: To make the students familiar with the basics of nursery raising.

Max. Marks: 75 **Course Credits** = 2+1=3

1.	Theory: 50	Written Exam: 45	Internal assessment: 5
2.	Practical: 25	Written Exam: 22	Internal assessment: 3

Periods per week: 1. Theory- Three of 45 minutes duration (Two Hours)
2. Practical- One of two hours duration (Two Hours)

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 9 short answer type questions of one mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 9 marks each and one question is to be attempted out of each unit.

Teaching Aids: Teachers can use lecture, lecture cum demonstration, project, assignments, problem solving, inductive and deductive methods, and can use L.C.D., L.E.D., Multimedia projector in addition to black/white board.

UNIT – I 8 Hours

Nursery and its importance: selection of nursery site, type of nursery, preparation of nursery bed. Soil solarisation and soil treatment.
Growing media: growing media, selection of media, quality of media, soil, sand, peat, leaf mould, saw dust, sphagnum moss, coco peat.
Scope and importance of vegetable and flower nurseries, mist chambers, polyhouses

UNIT – II 7 Hours

Sowing of seeds and planting material: Method of seed sowing, precautions taken during sowing & planting, application of manures and fertilizers, handling of plants.
Propagation, seed germination, dormancy and methods of breaking dormancy, pre-sowing seed treatment.
Seed hardening, seed priming, seed pelleting, irrigation systems, water soluble and liquid fertilizers, use of biocontrol agents, growth regulators, plant protection-techniques.
Transport and packing of nursery plants.

UNIT – III 8 Hours

Varieties and F_1 hybrids, seed rate, seed treatments, sowing, media-management for water, nutrients and plant protection, hardening, packaging, storage, transport and marketing of tomato, brinjal, chilli, capsicum, cucurbits, cabbage, cauliflower, carrot, radish, china aster, marigold, gaillardia, zinnia, petunia.

UNIT – IV **7 Hours**

Varieties and F_1 hybrids, seed rate, seed treatments, sowing, media-management for water, nutrients and plant protection, hardening, packaging, storage, transport and marketing of mango, citrus, lichi, lemon, apple, pear, peach, guava, forest trees like popular, eucalyptus.

GUIDELINES FOR PRACTICAL EXAMINATION

Time: 3 hrs **Max Marks: 25**
Practical: 22 **Internal Assessment: 03**

1. Demonstration of soil solarisation technique.
2. Preparation a raised nursery bed
3. Preparation of potting media.
4. Demonstration of seed sowing in plug-trays.
5. Demonstration of the potting of ornamental plants
6. Study of features of nursery-factors to be considered for setting up of nursery
7. Studying the problems of nursery business
8. Seed sowing-vegetable crops-annual flower crops
9. Study of Application of growth regulators for nursery
10. Irrigation and nutrient management-liquid fertilization
11. Study of care of nursery plants (shifting of plants, weeding, shade provision, watering, hardening).

Suggested Readings:

1. Grewal, H.S. 1999. *Propagation of ornamental plants*, Kalyani Publishers, New Delhi
2. Hartmann, H.T. and D.E. Kester; 1989. *Plant propagation-Principles and practices*. Prentice – Hall of India, New Delhi.
3. Nanda, K.K. and V.K. Kochhar; 1991. *Vegetative propagation and management*. Indus Publishing Company, New Delhi.
4. Sadhu, N.K. 1989. *Plant propagation*, Wiley Eastern Ltd., New Delhi.
5. Agarwal, R.L. 2005. *Seed Technology*. Oxford and IBH publishing Co. New Delhi.

B.A./B.Sc.(General) Agriculture Major/Minor

Course Title: BASICS OF AGRICULTURAL ECONOMICS

Course Code: (AGR-DSC-2 Maj/Min-201)
Semester: Second

Learning Outcome: To give student elementary information on basic principles of economics related to agriculture.

Max. Marks: 100		**Course Credits = 3+1=4**
1. Theory: 75	**Written Exam: 67**	**Internal Assessment: 8**
2. Practical: 25	**Written Exam: 22**	**Internal Assessment: 3**

Periods per week: 1. Theory- Four of 45 minutes duration (Three hours)
2. Practical- One of two hours duration (Two hours)

INSTRUCTIONS FOR PAPER SETTER AND EXAMINERS:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 10 short answer type questions of one and a half mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teachers can use lecture, lecture-cum-demonstration, project, assignments, problem solving, inductive and deductive methods and can use L.C.D., L.E.D., and multimedia projector in addition to black/white board.

UNIT – I **12 Hours**

Agricultural Economic: Meaning, scope and its relationship with other social and agricultural sciences.
Recourse Management in Agriculture: land,labour and capital.
Agriculture and Economic Development: Introduction, role, share of agriculture in economic development; Role of agriculture in economic development of India and its comparison with other countries.
Factor affecting agricultural development.

UNIT – II **11 Hours**

Risk and Uncertainty: Introductions, risk types and risk as choice indicator; different theories of risk and uncertainty; measures to minimize risk and uncertainty.
System of Farm Organization: Introductions, types, criteria for choice in India and suitability of alternate systems.
Farm Size and Practices: Holdings and farm.
Dynamics of Population: Population growth and its effect on agricultural growth rate.

UNIT – III **11 Hours**

Agricultural marketing, efficient marketing system, classification of markets, problems of Indian agricultural marketing and measure to improve agricultural marketing.
Rural Indebtness: introduction, extent, nature and causes of rural indebtness, Government measure to reduce rural indebtness
Agricultural Finance.
Agricultural credit institutions: structure of agriculture credit system, multi-agency approach, regional rural banks, NABARD, ARDC

UNIT – IV **11 Hours**

Self Help groups, cooperative credit structure-three tier structure
Rural Poverty: Introduction, factor responsible for poverty, suitable measures to reduce poverty and poverty alleviation programme.
Cottage and small industries: Features and classification of cottage industries. Difference between cottage and small industries, role and problems of cottage and small industries
Different measure to promote SSI and role of Government, small industries development bank of India and its role, khadi and village industries (KVIC)

GUIDELINES FOR PRACTICAL EXAMINATION

Time: 3 Hrs **Max Marks: 25**
Practical: 22 **Internal Assessment: 03**

1. Calculations of interests.
2. Calculations of depreciation.
3. Farm Budgeting
4. Preparations of balance sheet and other final statements of financial importance for agriculture farm.
5. Study tours of Marketing Centers of local area.

Suggested Readings:

1. Lekhi, R.K. and Singh, J., Agricultural Economics, Kalyani Publishers, Ludhiana, 2016.
2. Black. J.D., *Introduction of Economics for Agriculture,* Fromount Pierre National Press, 1955.
3. **Nanavati, M.B. and J. J. Anjaria, *The Indian Rural Problem*. The Indian Society of Agricultural Economics, Bombay, 1944.**
4. Memoria, C.B. and B.B., *Agricultural Problems in India,* Kitab Mahal, Allahabad, 2007.

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BIO-CHEMISTRY

NEP-2020COLLEGES SCIENCES TEMPLATE: STUDENTS OPTING FOR SINGLE MAJORAND SINGLE MINOR(TABLE1*)
AND STUDENTS OPTING FOR MULTIDICIPLINARY /INTERDICIPLINARY (TABLE2*)
B.Sc., B.Sc. HONOURS AND B.Sc. HONORS WITH RESEARCH IN BIOCHEMISTRY

Table 1*						Table2*					
Sem	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**	**Sem**	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**
I	L-100	(BCH-DSC-1 Maj/Min-101)	Biomolecules I	3+1=4	24	I	L-100	(BCH-DSC-1 Maj/Min-101)	Biomolecules I	3+1=4 (×3)	24
		(BCH-MDC-1 - 101)	Introduction to Biochemistry	2 +1=3				(BCH-MDC-1 - 101)	Introduction to Biochemistry	2 +1=3	
		VAC-1-101		2				VAC-1-101		2	
		*(BCH-SEC-1-101)	Biochemical technique-I	1 +2 = 3				*(BCH-SEC-1-101)	Biochemical technique-I	1 +2 = 3	
		(AEC-1-101)	English	2				(AEC-1-101)	English	2	
		(PUN-COMPL-1-101/HCP-COMPL-1-101	Pbi-1/HCP-1	2				(PUN-COMPL-1-101/HCP-COMPL-1-101	Pbi-1/HCP-1	2	
II	L-100	(BCH-DSC-2 Maj/Min-201)	Biomolecules II	3+1=4	24	II	L-100	(BCH-DSC-2 Maj/Min-201)	Biomolecules II	3+1=4 (×3)	24
		(BCH-MDC-2 - 201)	Fundamentals of Enzymology	2+1=3				(BCH-MDC-2 - 201)	Fundamentals of Enzymology	2+1=3	
		VAC-2-201		2				VAC-2-201		2	
		*(BCH-SEC-2-201)	Biochemical techniques II	1 +2 = 3				*(BCH-SEC-2-201)	Biochemical techniques II	1 +2 = 3	
		(AEC-2-201)	English -2	2				(AEC-2-201)	English -2	2	
		(PUN-COMPL-2-201/HCP-COMPL-2-201	Pbi-2/HCP-2	2				(PUN-COMPL-2-201/HCP-COMPL-2-201	Pbi-2/HCP-2	2	
THE CANDIDATE HAS TO SELECT MAJOR AND MINOR OUT OF 3 SUBJECTS STUDIED IN I AND II SEMESTER AT THE END OF THE SEMESTER II											
THE STUDENT EXITING THE PROGRAME AFTER ONE YEAR SECURING 48 CREDITS WILL BE AWARDED UG CERTIFICATE PROVIDED HE /SHE SECURE 4 CREDITS FROM SUMMER INTERSHIP /COMMUNITY WORK *Candidate has to select the SEC offered by the Department of Biochemistry in Colleges											

Notes:

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks inaggregate till semester 6 (upto 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc.HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s).To qualify for Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship.The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

BIO-CHEMISTRY
Semester I
Major/Minor

INSTRUCTIONS FOR PAPER SETTER AND STUDENTS:

1. Examiner will set total of NINE questions comprising TWO questions from each unit and ONE compulsory question.
2. All questions carry equal marks.
3. The compulsory question will consists of 12 short questions of one mark each covering the whole syllabi.
4. Besides Question No. 1, all other questions may contains 2-3 sub-parts.
5. Questions should be uniformly spread over the entire syllabus.
6. The students are required to attempt FIVE questions in all, ONE question from each unit and the Compulsory question.

(BCH-DSC-1 Maj/Min-101): Biomolecules-I (3+1)

Marks: 75 (60 + 15 Int.)

Objective: To learn about biomolecules, their structure and functional significance.

Learning Outcomes:

- To understand the chemical and molecular foundations of life and their roles in biological systems.
- To comprehend the classification, structure, chemical properties and biological functions of carbohydrates, and lipids.
- To perform independently the analysis of various biomolecules quantitatively in the laboratory.

SECTION-I

(Lectures: 10)

Introduction to biochemistry and its scope. Water: physical properties, as a biological solvent and structure of water, dissociation of water. pH and pOH, buffer solution. Henderson Hassel Balch equation, acid-base indicators, buffers and physiologically important buffers, dialysis and osmosis.

SECTION-II

(Lectures: 10)

Definition and classification of carbohydrates. Fischer and Haworth structures of carbohydrates. Stereoisomerism and mutarotation. Anomeric forms of monosaccharides. Derivatives of monosaccharides (glycosides, deoxy sugars, amino sugars and other derivatives of biological importance). Oligosaccharides of biological importance (structure of maltose, lactose, sucrose, cellobiose, trehalose, and raffinose). Characteristic reactions of monosaccharides: reactions with hydrazine, hydrogen cyanide, hydroxylamine; reduction and oxidation of sugars; Homo-and hetero-polysaccharides (structures of amylose, amylopectin, starch, glycogen, cellulose, chitin). Hetero-polysaccharides) as components of connective tissue. Polysaccharides of bacterial cell wall.

SECTION-III

(Lectures: 10)

Definition and classification of fatty acids (saturated and unsaturated). Essential fatty acids. Important reactions of functional groups present in fatty acids. Characteristics of fatty acids and fats (saponification, iodine, acid, acetyl and peroxide values). Refractive index, melting point, boiling point and their relation to molecular size. Properties of glycerol. Fats as source of energy. Waxes. Structures, characteristics and functions of lipids: triacylglycerol and phospholipids such as lecithin (Phosphatidylcholines), lysolecithins, cephalins (phosphatidylethanolamines), phosphatidylserines, Phosphatidylinositol, sphingomyelins, plasmalogens, cerebrosides, gangliosides, sulfatides.

SECTION-IV

(Lectures: 10)

Lipoproteins– composition, classification and biological functions. Liposomes. Terpenes and steroids –Terpenes of biological significance e.g., carotenes, phytol. Cholesterol and other animal sterols. Color reactions of sterols. Sterols of yeast and fungi (mycosterols). Phytosterols. Steroidal hormones. Bile acids. Structure and properties of eicosanoids-prostaglandins, leukotrienes, thromboxane, prostacyclins

PRACTICALS:

(BCH-DSC-1 Maj/Min-101): Biomolecules-I

Marks: 25 (20 +5 Int.)

1. Qualitative tests for carbohydrates.
2. Determination of achromatic point of saliva
3. Qualitative tests for Lipids/Fats.
4. Estimation of saponification value of fat/oil.
5. Estimation of acid value of fat/oil.
6. Isolation of starch from potato/gluten from wheat.
7. Demonstration of Osmosis and Dialysis

Suggested Books:

1. Lippincott's Illustrated Reviews: Biochemistry by P.C. Champe, R.A. Harvey& D.R. Ferrier, IndianEdition, 2017.
2. Harper's Illustrated Biochemistry by R.K. Murray, P.A. Mayes, D.K. Granner and V.W. Rodwell, 31th Ed. 2018, Prentice Hall International Ltd. (U.K.).
3. Voet D and Voet JG. Biochemistry 5th Edition, John Wiley and Sons Inc., 2021.
4. LehningerA.L.,NelsonD.L.andCox M.M.(2021)8th edition Principlesof biochemistry (W.H.Freeman,USA)
5. StryerL.,J.M.Berg,J.L.Tymoczko (2023)10th edition,Biochemistry(W.H.Freeman company, NewYork)
6. Conn E.E. and Stumpf P.K. (2006)5th edition Outlines of Biochemistry, (John Wiley and Sons, USA)
7. R.H. Garett and C.M. Grisham (2023)7th Edition Biochemistry,(Cengage Learning, US)
8. Zubay G.L., Parson W.W, Dennis E. Vance (William C. Brown, U.S.)1998 Principles of Biochemistry volume-1.

Semester II

Major/Minor

INSTRUCTIONS FOR PAPER SETTER AND STUDENTS:

1. Examiner will set total of NINE questions comprising TWO questions from each unit and ONE compulsory question.
2. All questions carry equal marks.
3. The compulsory question will consists of 12 short questions of one mark each covering the whole syllabi.
4. Besides Question No. 1, all other questions may contains 2-3 sub-parts.
5. Questions should be uniformly spread over the entire syllabus.
6. The students are required to attempt FIVE questions in all, ONE question from each unit and the Compulsory question.

(BCH-DSC-2 Maj/Min-201): Biomolecules-II (3+1) Marks: 75 (60 + 15 Int.)

Objective: To learn about biomolecules, their structure and functional significance.

Learning Outcomes:

- To comprehend the structure and functional properties of amino acids and proteins.
- To learn the structure and properties of Nucleic acids and porphyrins.
- To perform independently the analysis of various biomolecules quantitatively in the laboratory.

SECTION–I

(Lectures: 10)

Amino acids: common structural features. Stereoisomerism and RS system of designating optical isomers.
Classification based on the nature of "R" groups. Amino acids present in proteins and non-protein amino acids. Specialized role of amino acids. Physical and chemical properties of amino acids. Titration of amino acids. Rigid and planar nature of a peptide bond. Biologically active peptides.

SECTION–II

(Lectures: 10)

Proteins: Structural levels Primary, Secondary, Tertiary and Quaternary structures of proteins structure. Folding of peptide chains into regular repeating structures (α helix, β-pleated sheets). β -turn in polypeptides. Forces stabilizing structure and shape of proteins. Native proteins and their conformations. Behaviour of proteins in solutions. Salting in &salting out of proteins. Denaturation of proteins. Structural and functional diversity of proteins, fibrous proteins (keratins, collagen& elastin), globular proteins (hemoglobin, myoglobin) and conjugated proteins.

SECTION-III

(Lectures: 10)

Nucleic Acids: structure and properties of purine and pyrimidine bases. Nucleosides and nucleotides. Biologically important nucleotides. Double helical model of DNA and forces responsible for it. Different forms of DNA (A, B and Z type), denaturation of DNA. Physical and chemical properties of nucleic acids. Chemical and enzymatic hydrolysis of nucleic acids. Types of RNA and ribozyme.

SECTION-IV

(Lectures: 10)

Porphyrins: porphyrin nucleus and classification of porphyrins. Metalloproteins, Heme and other metalloporphyrins occurring in nature. Detection of porphyrins spectrophotometrically and by fluorescence. Chemical nature and physiological significance of bile pigments.

Suggested Books:

1. Lippincott's Illustrated Reviews: Biochemistry by P.C. Champe, R.A. Harvey & amp; D.R. Ferrier, Indian Edition, 2017.
2. Harper's Illustrated Biochemistry by R.K. Murray, P.A. Mayes, D.K. Granner and V.W. Rodwell, 31th Ed. 2018, Prentice Hall International Ltd. (U.K.).
3. Voet D and Voet JG. Biochemistry 5th Edition, John Wiley and Sons Inc., 2021.
4. Lehninger A.L., Nelson D.L. and Cox M.M. (2021)8th edition Principles of biochemistry (W.H. Freeman, USA)
5. StryerL., J.M. Berg, J.L. Tymoczko (2023)10th edition, Biochemistry (W.H. Freeman company, New York)
6. Conn E.E. and Stumpf P.K. (2006)5th edition Outlines of Biochemistry, (John Wiley and Sons, USA)
7. R.H. Garett and C.M. Grisham (2023)7th Edition Biochemistry, (Cengage Learning, US)
8. Zubay G.L., Parson W.W, Dennis E. Vance (William C. Brown, U.S.)1998Principles of Biochemistry volume-1

PRACTICALS:

(BCH-DSC-2 Maj/Min-201): Biomolecules-II

Marks: 25 (20 +5 Int.)

1. Qualitative tests for amino acids
2. estimation of DNA by Diphenylamine method
3. Separation of amino acids by thin layer chromatography
4. Ammonium sulphate fractionation of proteins.
5. Absorption Spectra of DNA and Proteins by UV visible spectroscopy
6. Estimation of Protein by Biuret /Lowry Method.

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BIOINFORMATICS

COURSE OUTLINE & SYLLABUS FOR BIOINFORMATICS

Students opting for Single major and Single minor(Table 1*)and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY(Table 2*)
B.Sc., B.Sc. Honours and B.Sc. Honours with Research (1st Year Programme)

Sem	Level	Paper Code	Paper Name	Credits (Th+Pr)
I	L-100	BINF-DSC-1 (Maj/Min-101)	Bioinformatics-I	3+1=4
		BINF-MDC-1-101	Introduction to Bioinformatics-I	2+1=3
		BINF-SEC-1-101	Tools for Biological Data Analysis	0+3=3
II	L-100	BINF-DSC-2 (Maj/Min-201)	Bioinformatics-II	3+1=4
		BINF-MDC-2-201	Introduction to Bioinformatics-II	2+1=3
		BINF-SEC-2-201	Computational Tools for Small and Macromolecules	0+3=3

Notes: -

1. MDC/IDC subject is offered by The Department of Bioinformatics for the students of allied/other streams. The students opting for Bioinformatics as DSC subject will select MDC/IDC from the list of courses offered by the allied/other departments.
2. Language paper chosen as MIL-1, MIL-2, MIL-3 and MIL-4 should be in the same subject.

SEMESTER - I

Paper: BINF-DSC-1-(Maj/Min-101): BIOINFORMATICS-I

External Theory Marks: 60
Internal Theory Marks: 15
Total Theory Marks: 75
Exam Duration:3 Hours

Course Duration:

1 Semester = 15 weeks which will have 45 hours of Theory Lectures (3 hours/week)and 30 hours of practical classes (2 hours/week)
1 Credit = 15 hours of theory lectures or 30 hours of practical classes
Total Credits: 4 (3 Credits for Theory Lectures and 1 Credit for Practical)
Each credit is equal to 25 marks

Objective:

Introduction to Biomolecules exposes the students to study structure and properties of biomolecules and their related bioinformatics resources.

General Instruction:

- The question paper will have seven questions, and each question have 12 marks. The first question would be compulsory having sub-parts covering the entire syllabus in the form of short and objective type questions.
- The remaining six questions will be set out of the Units I and II consisting of three questions from each unit.
- A candidate is required to attempt five questions in all by selecting two questions, from each unit and the first compulsory question.

UNIT – I

Nucleic acids:
Nitrogenous Bases, nucleosides and nucleotides.

Overview of DNA structure: Watson and Crick DNA double helix model, Conformations of DNA: A-DNA, B-DNA & Z-DNA.

Overview of RNA: types and roles of different RNAs: mRNA, rRNA, tRNA, snRNA and snoRNA.

Amino acids: Standard amino acids and the general properties.

Classification and structures: non-polar, polar but uncharged and charged amino acids.
Introduction to Proteins: Overview ofProtein Structure.
Primary structure - definition, peptide bond, dihedral angles, and Ramachandran plot.
Secondary structure - definition, alpha helix, beta sheet, and beta turn.
Tertiary structure - definition, interactions stabilizing tertiary structure.
QuaternaryStructure - definition, interactions stabilizing quaternary structure

UNIT – II

Introduction to Bioinformatics: Definitions, History, Goal, Scope and its Applications.

Literature Database: PubMed.

Biological Databases Classification: Primary, Secondary and Specialized databases.
Nucleotide Sequence Database: Genbank, EMBL, INSDC, Indian Biological Data Centre (IBDC) and its data portals.
Protein Sequence Database: Uniprot.
Database flat file formats: Genbank/Genpept, EMBL and Uniprot flat files.
Sequence file formats: FASTA

Sequence Alignment: Evolutionary basis of Alignment, Concept of Sequence Homology, Similarity, and Identity.Applications of Sequence Alignments.

Pairwise sequence alignment: Concept of Global and Local alignment, Dot Matrix method, Dynamic programming (Needleman Wunsch algorithm and Smith-Waterman algorithm) Concepts behind General gap and affine gap penalty.

Recommended Books:

1. Donald Voet and Judith G. Voet (2010) Biochemistry, 4th Edition, Wiley Publisher.
2. David L. Nelson and Michael M. Cox (2017). Lehninger Principles of Biochemistry, 7th Edition, WH Freeman Publisher.
3. Reginald H. Garrett and Charles M. Grisham (2023) Biochemistry, 7th Edition, Brooks/Cole Publisher.
4. Introduction to Bioinformatics by V. Kothekar, 1st Edition, (2006), Dhruv Publications, Delhi.
5. Bioinformatics by David Mount, Cold Spring Harbor Laboratory, (1989).
6. Bioinformartdics: A practical guide to the analysis of genes and proteins. Ed. By Baxevanis, 3rd Edition, (1998) John Wiley & Sons, Inc. Publication.

BINF-DSC-1-(Maj/Min-101):BIOINFORMATICS-I (PRACTICAL)

External Practical Marks:	20
Internal Practical Marks:	05
Total Practical Marks:	25

1. Draw the amino acids using Chemsketch.
2. Plotting of Ramachandran map of a protein.
3. Visualization of DNA using Pymol.
4. Searching literatures over PubMed.
5. Understanding and using Sequence information resources on web: EMBL-ENA,Genbank, EMBL, IBDC and Uniprot.
6. The information retrieval using Gquery.
7. Performing Sequence alignment using DOT PLOT method

SEMESTER - II

Paper: BINF-DSC-2-(Maj/Min-201): BIOINFORMATICS-II

External Theory Marks: 60
Internal Theory Marks: 15
Total Theory Marks: 75
Exam Duration:3 Hours

Course Duration:

1 Semester = 15 weeks which will have 45 hours of Theory Lectures (3 hours/week)and 30 hours of practical classes (2 hours/week)
1 Credit = 15 hours of theory lectures or 30 hours of practical classes
Total Credits: 4 (3 Credits for Theory Lectures and 1 Credit for Practical)
Each credit is equal to 25 marks

Objective:

This subject offers various bioinformatics tools that helps in analysis of biological sequence data and phylogenetic analysis of proteins and nucleotides by using various computational tools.

General Instruction:

- The question paper will have seven questions, and each question have 12 marks. The first question would be compulsory having sub-parts covering the entire syllabus in the form of short and objective type questions.
- The remaining six questions will be set out of the Units I and II consisting of three questions from each unit.
- A candidate is required to attempt five questions in all by selecting two questions, from each unit and the first compulsory question.

UNIT – I

Tools for Pairwise Sequence Alignment:BLAST: Algorithm, types, understanding the BLAST output and the significance of E-value and Bit score, different uses of BLAST.

FASTA algorithm, Difference between BLAST & FASTA.

Substitution matrices: Amino acid substitution matrices (PAM and BLOSUM)

Multiple Sequence Alignment (MSA): SP (Sum of Pairs) measure to determine scoring function, Multiple Sequence Alignment methods: Progressive (Clustal Omega and PILEUP), Iterative and Block-based methods of MSA. Application of MSA.

Tools for MSA: Clustal Omega, Clustal-X and T-coffee.

UNIT – II

Phylogenetic Analysis: Basic terminology in Phylogenetics, Introduction to phylogenetic analysis and its application, Phylogenetic tree terminology and topologies.

Methods of Phylogenetic Tree Construction:
Distance Based Methods: NJ, FM & UPGMA
Character Based Methods: Maximum Parsimony & Maximum Likelihood.
Phylogenetic Tree Evaluation Methods: Bootstrapping & Jackknifing.

Tools for Phylogenetic Analysis: Introduction to Phylip and MEGA packages.

Protein Structural databases:PDB, SCOP and CATH

Recommended Books:

- Bioinformatics: A modern approach by Vittal R. Srinivas, Prentice/ Hall of India Pvt. Ltd. New Delhi, (2005)
- Bioinformatics by David Mount, Cold Spring Harbor Laboratory, (1989).
- Bioinformatics: Ardpractical guide to the analysis of genes and proteins. Ed. By Baxevanis 3rd Edition, (1998) John Wiley & Sons, Inc. Publication
- BLAST by Ian Korf, Mark Yandell, Joseph Bedell, O'Reilly, 1st Edition, (2003).
- Molecular Evolution: a phylogenetic approach, (1998), 352p, ROM and Holmas FC, Blackwell Science.

BINF-DSC-2-(Maj/Min-201):BIOINFORMATICS-II (PRACTICAL)

External Practical Marks:	20
Internal Practical Marks:	05
Total Practical Marks:	25

1. Performing BLASTp/n and interpreting its results.
2. Performing Multiple Sequence Alignment using Clustal Omega
3. Building Phylogenetic tree using:
   a) Distance-based & character-based methods by PHYLIP or MEGA
   b) Tree evaluation
4. Understanding the PDB and its file format
5. Downloading and visualizing protein structure from PDB

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BIOTECHNOLOGY

NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR(Table 1*)
and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY(Table 2*)
B.Sc., B.Sc. HONOURS AND B.Sc. HONOURS WITH RESEARCH

Table 1*						*Table 2**					
Sem	Level	Course Code	Course Name	Credits (Th+Pr+Tut)	Total Credits	Sem	Level	Course Code	Course Name	Credits (Th+Pr+Tut)	*Total Credits*
I	L-100	BIOT-DSC-(MAJ/MIN)-1-101	Introduction to Biotechnology	3+1	24	I	L-100	BIOT-DSC-1 (MAJ/MIN)-101	Introduction to Biotechnology	3 + 1	*24*
		*MIN B-1		3+1				Major		3+1	
		*MIN C-1		3+1				Major		3+1	
		BIOT-MDC-1- 101	Foundations of Biotechnology-I	2+1				BIOT-MDC1- 101	MDC	2 + 1	
		**BIOT-SEC-1-101	Biosafety & Biosecurity	2+1				VAC-1-101	Value-Added Course	2	
		BIOT-VAC-1-101	Value Added Course	2				SEC-1-101	SEC	2+1	
		AEC-1-101	Eng-1	2				AEC-1-101	Eng-1	2	
		BIOT-PUN (Comp-1)-101 BIOT-HCP (Comp-1)-101	Pbi/HCP	2				PUN (Comp-1)-101 HCP (Comp-1)-101	Pbi/HCP	2	
II	L-100	BIOT-DSC-(MAJ/MIN)-2-201	Foundations of Microbiology	3+1	24	II	L-100	BIOT-DSC-2 (MAJ/MIN)-201	Foundations of Microbiology	3 + 1	*24*
		*MIN-B-2		3+1				Major		3+1	
		*MIN-C-2		3+1				Major		3+1	
		BIOT-MDC-2- 201	Foundations of Biotechnology-II	2+1				MDC-2-201	MDC	2 + 1	
		**BIOT-SEC-2-201	Biostatistics	2+1				VAC-2-201	Value-Added Course	2	
		AEC-2-201	Eng-2	2				SEC-2-201	SEC	2+1	
		BIOT-PUN (Comp-2)-201 BIOT-HCP (Comp-2)-201	Pbi/HCP	2				AEC-2-201	Eng-2	2	
		BIOT-VAC-2-201	Value Added Course	2				PUN (Comp-2)-201 HCP (Comp-2)-201	Pbi/HCP	2	

The candidate has to select Major and Minor out of 3 subjects studied in the I and II Semester at the end of the II Semester

The student exiting the programme after oneyear, securing 48 credits, will be awarded UG Certificate provided he/she secures 4credits from summer internship/community work.

Semester I

BIOT-DSC-(MAJ/MIN)-1-101: INTRODUCTION TO BIOTECHNOLOGY

Learning Outcomes: This course will introduce the basic concepts and skills of biotechnology to the students. They will learn about the history of biotechnology; the foundations of modern biotechnology; the role of biotechnology in industry, environment and modern medicine and the ethical implications of biotechnology.

Course Duration

45 hours of lectures for Theory + 30 hours of practical
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)

Credits : 4 credits (Each credit is equal to 25 marks)

External Theory	Internal Theory	External practical	Internal practical
68	07	22	03

Theory Paper: The exam of this course will be of 3 hours duration.

Instructions for paper setters and candidates:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. The students are required to attempt one question from each unit.

Compulsory question will be of 12 marks. Rest all questions will be of 14 marks each

UNIT I

Advent, scope and basics of biotechnology
Bacteria as work horses of biotechnology; *E.coli* as the model bacterium
Saccharomyces cerevisiae in Biotechnology
Introduction of multicellular organisms as research models:*Drosophila melanogaster, Caenorhabditiselegans, Daniorerio, Musmusculus,, Arabidopsis thaliana* as model for plant genetics.

UNIT II

Genome organization in prokaryotes and eukaryotes, Molecular nature of the genetic material.
Introduction of concept and applications of genomics, transcriptomics, proteomics and metabolomics.
Introduction to Principle and basic techniques like sterilization, centrifugation& Agarose gel elcrtrophoresis.

UNIT III

Applications of biotechnology: Green technology to controlpollution.
Role of biotechnology in medicine and diagnostics, agriculture and industry. Introduction of gene therapy.

UNIT IV

Genetically modified organisms (GMOs)-transgenic plants and animals and their applications in biotechnology. Bioterrorism and biowarfare. Ethical, Social and Legal implications of Biotechnology.

Reference Books:

1. Comprehensive Biotechnology, Murray Moo-Young, 2ndedition(2011), Pergamon Press.
2. Introduction to Biotechnology ,William J. Thieman& Michael A. Palladino, 3rd edition (2012),Benjamin Cummings.
3. Biotechnology Expanding Horizons, B D Singh,4thedition(2012),Kalyani Publishers.
4. The Ethics of Biotechnology (Biotechnology in the21st Century), Jonathan Morris, Chelsea HousePub (L),1stedition(2005).
5. Biotechnology, Applying the Genetic Revolution, David P.Clark& Nanette J.Pazdernik,1stedition(2008), Academic Cell.
6. Molecular Biotechnology, Sandy B.Primrose, 2ndedition(1991), Blackwell Scientific Publications,
7. Biotechnology:Demystifying the concepts, David Bourgaize,Thomas R. Jewell & Rodolfo G. Buiser,1st Edition(1999), Benjamin Cummings.
8. Ethical Issues in Biotechnology, Richard Sherlock & John D. Morrey, 1st edition (2002), Rowman& Little field Publishers.

BIOT-DSC-(MAJ/MIN)-1-101: INTRODUCTION TO BIOTECHNOLOGY(Practicals)

1. Introduction and use of Auto-pipettes, pH meter, spectrophotmeter
2. Use of centrifuges, light microscope, electrophoretic apparatus
3. Introduction to vortex mixer, magnetic stirrer, rocker, laminar hoods, autoclave
4. DNA isolation
5. Restriction digestion
6. Cell lysis of bacterial cell (Lysozyme digestion)

Semester II

BIOT-DSC-(MAJ/MIN)-2-201: FOUNDATIONS OF MICROBIOLOGY

Learning Outcomes: This paper exploresthe interactions of the Microbes with environment and other living organisms. This will enhance the student's understanding about how the microbes play a significant role in medical science and industries. The study of microbesfrom basic to advance level, with an understanding of biochemistry,cell structure, and application makesthis paper significant.

Course Duration

45 hours of lectures for Theory + 30 hours of practical
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)

Credits: 4 credits (Each credit is equal to 25 marks

External Theory	Internal Theory	External practical	Internal practical
68	07	22	03

Theory Paper: The exam of this course will be of 3 hours duration.

Instructions for paper setters and candidates:

The syllabus of this paper has been divided into FOUR units.
Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. The students are required to attempt one question from each unit.

Compulsory question will be of 12 marks. Rest all questions will be of 14 marks each

UNIT I

History of Microbiology: A. Leewenhook, L. Pasteur, R. Koch, J. Lister, J. Tyndall,Alexander Flemming, Koch postulates.
Principle and applications of microscopy: Bright field, dark field, phase contrast, fluorescent, electronmicroscopy.

UNIT II

Microbial classification; Conventional and Modern techniques.Morphology of bacteria with major emphasis on bacterial structure especially cell wall.Gram-positive and Gram-negative bacteria. Microbial spores.

UNIT III

Microbial growth, phases of growth, generation time, growth rate, monoauxic, diauxic andsynchronous growth. Chemostat and turbidostat.
Physical and chemical agents to kill microbes major emphasis on sterilization process.

UNIT IV

Normal micro flora in human. Types of microbial pathogens and diseases caused by them. Microbial interactions like symbiosis and antibiosis.

Reference Books:

1. Cappuccino J.G., Sherman N. (2007). Microbiology: A laboratory manual, (PearsonBenjamin Cummings).
2. Tortora, G.J., Funke, B.R. and Case, C.L. (2009) Microbiology: An introduction(Benjamin/ Cummings publishingcompany, Inc).
3. R. Y. Stanier, M. Doudoroff, E. A. Adelberg (1999). General microbiology (MacMillianpress London).
4. M.J. Pelczar, E.C. Sun Chan, N.R. Krieg (1986). Microbiology (Tata McGraw HillPublication, New
Delhi).
5. H.G. Schlegel, C.Zaborosch, M. Kogut (1993).General microbiology (CambridgeUniversity Press).
6. S.C. Prescott, C.G. Dunn (1959). Industrial microbiology (McGraw- Hill).
7. Purohit, S.S. (2003). Microbiology: Fundamentals and applications (Agrobios, India).
8. Postgate, J.R. (2000). Microbes and man (Cambridge University Press).

BIOT-DSC-(MAJ/MIN)-2-201: FOUNDATIONS OF MICROBIOLOGY (Practicals)

1. Cleaning of glass wares, preparation of media, cotton plugging and sterilization.
2. Isolation of microorganisms from air, water and soil samples.
3. Dilution and pour plating techniques.
4. Gram staining
5. Growth curve of microorganisms.

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BOTANY

NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR<u>SINGLE MAJOR AND</u><u>SINGLE MINOR</u>(Table 1*) and STUDENTS OPTING FOR<u>MULTI-DISCIPLINARY/ INTER-DISCIPLINARY</u>(Table 2*)<u>B.Sc., B.Sc. HONOURS AND B.Sc. HONOURS WITH RESEARCH IN BOTANY</u>

Table1*						Table2*					
Sem	**Level**	**Course Code**	**CourseName**	**Credits(Th+Pr+Tut)**	**Total Credits**	**Sem**	**Level**	**Course Code**	**CourseName**	**Credits(Th+Pr+Tut)**	**Total Credits**
I	L-100	(BOT-DSC-1 Maj/Min-101)	Plant DiversityI	3+1=4(x3)	24	I	L-100	(BOT-DSC-1 Maj/Min-101)	PlantDiversityI	3+1=4(x3)	24
		(BOT-MDC-1-101)	Plants–Importance andPropagation	3				(BOT-MDC-1-101)	Plants–Importance andPropagation	3	
		(BOT-VAC-1-101)	Nurseryand Gardening	2				(BOT-VAC-1-101)	Nurseryand Gardening	2	
		(BOT-SEC-1-101)	BasicLabandField Skills inBotany	2+1=3				(BOT-SEC-1-101)	BasicLab andFieldSkills in Botany	2+1=3	
		(AEC-1-101)	Eng-1	2				(AEC-1-101)	Eng-1	2	
		(PUN-COMPL-1-101/HCP-COMPL-1-101)	Pbi-1/HCP-1	2				(PUN-COMPL-1-101/HCP-COMPL-1-101)	Pbi-1/HCP-1	2	
II	L-100	(BOT-DSC-2 Maj/Min-201)	PlantDiversityII	3+1=4(x3)	24	II	L-100	(BOT-DSC-2 Maj/Min-201)	PlantDiversityII	3+1=4(x3)	24
		(BOT-MDC-2-201)	Utilizationand Conservationof Plants	3				(BOT-MDC-2-201)	Utilizationand Conservation ofPlants	3	
		(BOT-VAC-2-201)	Biodiversityand Conservation	2				(BOT-VAC-2-201)	Biodiversityand Conservation	2	
		(BOT-SEC-2-201)	Biofertilizers	2+1=3				(BOT-SEC-2-201)	Biofertilizers	2+1=3	
		(AEC-2-201)	Eng-2	2				(AEC-2-201)	Eng-2	2	
		(PUN-COMPL-2-201/HCP-COMPL-2-201)	Pbi-2/HCP-2	2				(PUN-COMPL-2-201/HCP-COMPL-2-201)	Pbi-2/HCP-2	2	
The candidate has to select Major and Minor out of 3 subjects studied in the I and IISemester at the end of the SemesterII											
The student exiting the programme after one year, securing 48 credits,will be awarded UG Certificate provided he/ she secures 4 credits from Summer internship/ community work.											

***Note:The candidate will opt for either Template 1or 2 from Semester III onwards.**

Notes:

1. In the first semester student is offered three different subjects A, B& C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjectsand complete his/her three-year degree by

securing 36 credits each in the chosen two subjectsasMajorand24creditsinMinorsubject.

4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% creditsof the total credits in theMajor subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject ,the student must attain atleast 12 credits inthat subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rdsemester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of two Major subjects.

CourseTitle: PLANT DIVERSITY-I **CourseCode:BOT-DSC-1**
Maj/Min-101

Learning Outcome: The basic objective of this paper is to make students aware of the diversity in life forms of plant kingdom. A systematic study of algae, fungi and bryophytes would familiarize students with the structural differentiation andheterotrophi cand autotrophic modes of nutrition in plant kingdom.This paper will form the basis of advanced study in Botany.

	Max.Marks:100		**CourseCredits**=	3+1=4
1.	Theory:75	WrittenExam:67	InternalAssessment:	8
2.	Practical:25	WrittenExam:22	InternalAssessment:	3

Periods per week: 1.Theory-Four of 45 minutes duration (Threehours)
2.Practical-One of two hours duration

Instructions for Paper Setters:

Question Paper will have four units.Students will attempt five questions out oftotal nine comprising of one COMPULSORYquestion consisting of 15 questions having 8 MCQ and 7 fill in the blanks of one mark each covering the whole syllabus.Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teachers can use lecture, lecture-cum-demonstration, project, assignments, problem solving, inductive and deductive methods and can use L.C.D., L.E.D., and multimedia projector in addition to black/white board.

UNIT–I **12Hours**

Algae: General Characters; systematic position, structure and life history of *Oscillatoria* (Cyanophyceae) *Volvox* (Chlorophyceae), *Ectocarpu*s (Phaeophyceae), *Batracospermum* (Rhodophyceae) and economic importance of algae.

UNIT–II **11Hours**

Fungi: General characters; systematic position, structure and life history of *Albugo candida* (White rust of crucifers), *Rhizopus* and *Saccharomyces*.

UNIT–III **11Hours**

Systematic position, structure and life history of *Ustilago tritici* (Smut of wheat),*Colletotrichum falcatum* (Red rot of sugarcane).
SymbioticAssociation:General account of Lichens and their significance.

UNIT–IV **11Hours**

Bryophyta: General characters; systematic position, structure, reproduction and life cycle of *Riccia* (Hepaticopsida); *Anthoceros* (Anthocerotopsida) and *Funaria* (Bryopsida) excluding developmental stages.

Suggested Readings:

1. Alexopoulos,C.J.Mims,C.W`.and Blackwell ,M.1996. Introductory Mycology. John Wiley and Sons, Inc. USA.
2. Dube,H.C.1990.An Introduction to Fungi.Vikas Publishing House Pvt.Ltd.,Delhi.
3. Goyal, J.P.Dhand, Neelamand Saini, Aruna.2011. Foundation sof Botany.Trueman Book Company Jalandhar.
4. Pandey, B.P.2009.Botany for Degree Students. S.Chand & Co.Ltd., NewDelhi.
5. Sharma, O.P.1992 .Text book of Thallophytes, McMillanIndia Ltd., NewDelhi.
6. Sharma,P.D.1991.The Fungi .Rastogi & Co., Meerut.
7. Smith,G.M.1971.Cryptogamic Botany .Vol .1, Algae & Fungi, Tata McGraw Hill Publishing Co. New Delhi.
8. Singh, V. Pandey, P.C.and Jain, D.K.2012.Text book of Botany, Diversity of Microbes and Cryptogams. Rastogi Publications, Meerut & New Delhi.
9. Srivastava, H.N.2013. Pradeep's Botany Vol.I (Diversity of Micro be sand Cryptogams), Pradeep Publications, Jalandhar (India).
10. Vashishta, B.R.1999. Botany for Degree Students. Algae.S. Chandand Company Ltd., New Delhi.
11. Vashishta,B.R.1999.Botanyf or Degree Students. Fungi.S.Chandand Company Ltd., New Delhi.
12. Puri,P.1980. Bryophyta. AtmaRam & Sons ,Delhi.
13. Vashishta, B. R. 1999. Botany for Degree Students. Bryophyta. S. Chand and Company Ltd., New Delhi

Suggested laboratory exercises:

1. Study of morphology of various genera included in algaeand fungi.
2. Study of Crustose, Folio seand Fructico setypes of Lichenthalli.
3. Histopathological study of Whiterus to fcrucifers, Smut of wheat and Red-rot of sugarcane.
4. *Riccia*: Study of morphology of thallus, Study of L.S. Mature sporogonium (through permanent slides) and Preparation of permanent stained slide of V.S.*Riccia* thallus.
5. *Anthoceros*: Study of morphology of thallus, Study of T.S. Thallus passing through antheridia, T.S. Thallus passing through archegonia, L.S. Mature sporogonium. (through permanent slides) and Preparation of permanent stained slide of V.S. thallus.
6. *Funaria*:Study of morphology, Study of L.S.Malereceptacle, L.S.Femalereceptacle, L.S.Capsule (through permanent slides) and Preparation of permanent stained slide of T.S.of stem.

GUIDELINES FOR PRACTICAL EXAMINATION

Time-3Hrs **MaxMarks:25**

Practical:22 **Internal Assessment:03**

1. Identify and write illustrated morphological notes on specimen A and B. (06)
2. Cut T.S., stain and make a permanent mount of specimen C (from bryophytes). Identify and draw its labeled diagramand show slide to examiner. (06)
3. Identify the slides Dand Egiving atleast one reas on for each. (04)
4. Practical Note-book (03)
5. Viva-voce. (03)

Course Title: PLANT DIVERSITY-II **CourseCode:BOT-DSC-2**
Maj/Min-201

Learning Outcome: The basic objective of this paper is to make the students aware of the diversity in various life forms of plant kingdom. A sequential study ranging from Pteridophytes - the first vascular land plants to gymnosperms, would enablestudents to have a broad prospective of evolutionary trends in plant kingdom**.**

Max.Marks:100		**Course Credits**=	3+1=4
1. Theory:75	Written Exam:67	Internal Assessment:	8
2. Practical:25	Written Exam:22	Internal Assessment:	3

Periods per week:1.Theory- Four of 45 minutes duration (Three Hours)
2. Practical- One of two hours duration (TwoHours)

Instructions for Paper Setters:
Question Paper will have four units.Students will attempt five questions out of total nine comprising of one COMPULSORY question consisting of 15 questions having 8 MCQ and 7 fill in the blanks of one mark each covering the whole syllabus.Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teachers can use lecture, lecture-cum-demonstration, project, assignments, problem solving, inductive and deductive methods and can use L.C.D., L.E.D., and multimedia projector in addition to black/white board.

UNIT–I **12Hours**

Pteridophyta: General characters; systematic position, structure and reproduction of *Rhynia* (Psilophytopsida) and *Selaginella* (Lycopsida) excluding developmental stages, heterosporyand seed habit.

UNIT–II **11Hours**

Systematic position, structure and reproduction of *Equisetum* (Sphenopsida) and *Pteris* (Pteropsida) excluding developmental stages.

UNIT–III **11Hours**

Gymnosperms: General characters and economic importance of gymnosperms; systematic position, structure and reproduction of *Cycas* excluding developmental stages.

UNIT–IV **11Hours**

Systematic position, structure and reproduction of *Pinus* and *Ephedra* excludingdevelopmental stages.

Suggested Readings:

1. Goyal, J.P. Dhand, Neelamand Saini, Aruna 2011. Foundations of Botany. Trueman Book Company Jalandhar.
2. Pandey, B.P.2009. Botany for Degree Students.S. Chand & Co.Ltd., NewDelhi
3. Sharma, O.P. 1990. Textbook of Pteridophyta, Mc Graw HillPublishingCo., NewDelhi.
4. Singh, V.Pandey, P.C. and Jain, D.K. 2012. Textbook of Botany,Diversity ofMicrobes and Cryptogams. Rastogi Publications, Meerut & New Delhi.
5. Smith,G.M.1971.CryptogamicBotany.Vol.II,BryophytesandPteridophytes,TataMcGraw Hill Publishing Co., New Delhi.
6. Srivastava, H.N. 2013.Pradeep's Botany Vol. I (Diversity of Microbes and Cryptogams),Pradeep Publications, Jalandhar (India).
7. Bhatnagar, S.P. and Moitra, A. (1996) *Gymnosperms*. New Age International (P) Ltd. Publishers, New Delhi, India.
8. Chopra,G.L.TextbookofGymnosperms.S.Nagin.Delhi.1976
9. Sporne, K.R. The morphology of Gymnosperms, Hutichnson & Co. (Publishers) Ltd, London 1965.

Suggestedlaboratoryexercises:

1. *Selaginella*: Study of morphology, study of L.S. sporangiferous spike (through permanent slide) and preparation of permanent slide of T.S. stem.
2. *Equisetum*: Study of morphology, study of L.S. strobilus and T.S. strobilus (through permanent slide) and preparation of permanent slide of T.S. Aerial stem passing through internode.
3. *Pteris*: Study of morphology, study of mature prothallus (through permanent slide) and preparation of permanent slide of T.S. petiole.
4. *Cycas*: Morphology (coralloid root, scale and foliage leaves, male cone, megasporophyll). Preparation of permanent slide of *Cycas* (T.S. leaflet)
5. *Pinus*: Morphology (dwarf shoot, male and female cone and seed), preparation ofpermanent slide of *Pinus* (T.S. Needle).
6. *Ephedra*:Morphology(maleandfemalecone),preparationofpermanentslideof *Ephedra* (T.S. Stem).

GUIDELINES FOR PRACTICAL EXAMINATION

Time:3Hrs **MaxMarks:25**

Practical:22 **InternalAssessment:03**

1. IdentifyandwriteillustratedmorphologicalnoteonspecimenAandB. (06)
2. Cut T.S., stain and make a permanent mount of specimen C. Identify and draw its labelled diagramandshowslidetoexaminer. (06)
3. IdentifytheslidesDandEgivingatleastonereasonforeach. (04)
4. PracticalNote-book (03)
5. Viva-Voce. (03)

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CHEMISTRY

CURRICULUM AND CREDIT DISTRIBUTION FOR CHEMISTRY AS MAJOR SUBJECT AS PER TABLE 1 AND TABLE 2

Table 1						Table 2					
Sem	**Level**	**Course Code**	**Course Name**	**Credits**	**Total Credits**	**Sem**	**Level**	**Course Code**	**Course Name**	**Credits**	**Total Credits**
I	L-100	(CHM-DSC-1 MAJ/MIN-101)	GENERAL CHEMISTRY – 1	4	24	I	L-100	(CHM-DSC-1 MAJ/MIN-101)	GENERAL CHEMISTRY – 1	4	24
		B1	To be chosen by the student	4				B1	To be chosen by the student	4	
		C1	To be chosen by the student	4				C1	To be chosen by the student	4	
		IDC-1	To be chosen from the available options	3				IDC-1	To be chosen from the available options	3	
		(CHM-SEC-1-101)	Hands on chemistry Lab skills-1/ to be chosen from available options	3				(CHM-SEC-1-101)	Hands on chemistry Lab skills-1/ to be chosen from available options	3	
		ENG-1	Compulsory	2				ENG-1	Compulsory	2	
		VAC-1	To be chosen from the common basket	2				VAC-1	To be chosen from the common basket	2	
		PBI-1/HCP-1	Compulsory	2				PBI-1/HCP-1	Compulsory	2	
II	L-100	(CHM-DSC-2 MAJ/MIN-201)	GENERAL CHEMISTRY – 2	4	24	II	L-100	(CHM-DSC-2 MAJ/MIN-201)	GENERAL CHEMISTRY – 2	4	24
		B2	To be chosen by the student	4				B2	To be chosen by the student	4	
		C2	To be chosen by the student	4				C2	To be chosen by the student	4	
		IDC-2	To be chosen from the available options	3				IDC-2	To be chosen from the available options	3	
		(CHM-SEC-2-201)	Hands on chemistry Lab skills-2/	3				(CHM-SEC-2-201)	Hands on chemistry Lab skills-	3	

			to be chosen from available options						2/ to be chosen from available options		
		ENG-2	Compulsory	2				ENG-2	Compulsory	2	
		VAC-2	To be chosen from the common basket	2				VAC-2	To be chosen from the common basket	2	
		PBI-2/HCP-2	Compulsory	2				PBI-2/HCP-2	Compulsory	2	
III	L-200	(CHM-DSC-3 MAJ/MIN-301)	GENERAL CHEMISTRY – 3	4	24	III	L-200	(CHM-DSC-3 MAJ/MIN-301)	GENERAL CHEMISTRY – 3	4	24
		(CHM-DSC-4 MAJ-302)	BASIC CONCEPTS IN CHEMISTRY - 1	4				Major - 3	To be chosen by the student	4	
		Minor – 3	To be chosen by the student	4				Minor – 3	To be chosen by the student	4	
		Minor - 4	To be chosen by the student	4				Minor - 4	To be chosen by the student	4	
		IDC-3	To be chosen from the available options	3				IDC-3	To be chosen from the available options	3	
		(CHM-SEC-3-301)	Hands on chemistry Lab skills-3/ to be chosen from available options	3				(CHM-SEC-3-301)	Hands on chemistry Lab skills-3/ to be chosen from available options	3	
		ENG-3	Compulsory	2				ENG-3	Compulsory	2	
IV	L-200	(CHM-DSC-5 MAJ-401/MIN-302)	GENERAL CHEMISTRY – 4	4	24	IV	L-200	(CHM-DSC-5 MAJ-401/MIN-302)	GENERAL CHEMISTRY – 4	4	24
		(CHM-DSC-6 MAJ-402/MIN-401)	GENERAL CHEMISTRY – 5	4				(CHM-DSC-6 MAJ-402/MIN-401)	GENERAL CHEMISTRY – 5	4	
		(CHM-DSC-7 MAJ-403)	BASIC CONCEPTS IN CHEMISTRY - 2	4				Major - 5	To be chosen by the student	4	
		(CHM-	BASIC	4							

		DSC-8 MAJ-404)	CONCEPTS IN CHEMISTRY - 3					Major - 6	To be chosen by the student	4
		Minor - 5	To be chosen by the student	4				Minor - 5	To be chosen by the student	4
		ENG-4	Compulsory	2				ENG-4	Compulsory	2
		VAC-3	To be chosen from the common basket	2				VAC-3	To be chosen from the common basket	2

CURRICULUM AND CREDIT DISTRIBUTION FOR CHEMISTRY AS MINOR SUBJECT AS PER TABLE 1 AND TABLE 2

Table 1					Table 2				
Sem	**Level**	**Course Code**	**Course Name**	**Credits**	**Sem**	**Level**	**Course Code**	**Course Name**	**Credits**
I	L-100	(CHM-DSC-1 MAJ/MIN-101)	GENERAL CHEMISTRY – 1	4	I	L-100	(CHM-DSC-1 MAJ/MIN-101)	GENERAL CHEMISTRY – 1	4
II	L-100	(CHM-DSC-2 MAJ/MIN-201)	GENERAL CHEMISTRY – 2	4	II	L-200	(CHM-DSC-2 MAJ/MIN-201)	GENERAL CHEMISTRY – 2	4

CURRICULUM AND CREDIT DISTRIBUTION FOR CHEMISTRY AS IDC AS PER TABLE 1 AND TABLE 2

Table 1					Table 2				
Sem	**Level**	**Course Code**	**Course Name**	**Credits**	**Sem**	**Level**	**Course Code**	**Course Name**	**Credits**
I	L-100	(CHM-IDC-1-101)	Basic Analytical Methods-1	3	I	L-100	(CHM-IDC-1-101)	Basic Analytical Methods-1	3
II	L-100	(CHM-IDC-2-201)	Basic Analytical Methods-2	3	II	L-100	(CHM-IDC-2-201)	Basic Analytical Methods-2	3

Note:-

1. In the first semester, studentsare offered three different subjects: A, B & C.
2. At the end of the second semester, each student has the option to choose any one of the two streams, namely Single Major or Multi-Disciplinary. The student choosing the Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student choosing the Multi-Disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in the Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for a 4-year degree B.SC HONOURS with research, on the condition that the Institution is a PU-approved Research Centre
5. The student opting for a Single Major must secure 50% of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for a Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year, securing 48 credits, will be awarded a UG Certificate provided he/she secure 4 credits from summer internship/community work. Further, the student exiting the programme after two years, securing 96 credits, will be awarded a UG Diploma provided he/she secure 4 credits from a summer internship. The student successfully securing 144 credits will be awardeda BA/BSc degree.
7. Language papers chosen as MIL-1 and MIL-2 should be on the same subject.
8. Once Major and Minor papers are chosen in the 3rd semester, the same will continue as Major and Minor subjects in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write a Dissertation of 08 credits in one of the two Major subjects.

Semester	**I**
Code of Paper	**CHM-DSC-1-MAJ/MIN-101**
Title of Paper	**General Chemistry-1**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100(75Theory + 25 Practical)**
Marks distribution of the Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of the Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. per week** **Practical: 2 Hrs. per week**

Applicable to the students

1. **Those who have opted for chemistry as a Major subject in Table 1 or Table 2**
2. **Those who have opted for chemistry asa Minor subject in Table 1 or Table 2**

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

By studying this course, the students will be able to:

- Explain the electronic configuration of elements & ions based on different principles and rules.
- Explain the chemical behaviour of different elements from their position in the periodic table.
- Explain the differential behaviour of organic compounds based on fundamental concepts learnt.
- Explain and differentiate between various types of reagents, organic reactions, reactive intermediates and how to assign a formal charge to an ionic species.
- The student shall also be able to differentiate between geometrical and conformational isomers.
- Explain the factors influencing the rate of reactions based on various theories of chemical kinetics.
- Explain the concept of the order of a reaction by different methods.
- Understand different types of catalysts and the mechanism of enzyme catalysis.
- Emphasis is given to developing practical skills that will aid the student to set up accurate and reproducible, result-oriented experiments in higher classes.

GENERIC OUTCOMES

The first year of the course is designedto inculcate a scientific temper amongst the students. This is done by teaching them how to collect samples/data/evidence, carry out analysis, evaluation and draw logical conclusions.To hone their critical thinking and enhance creativity, which shall lead to innovative and out-of-the-box thinking, which shall help in complex problem-solving situations in higher classes.They are also encouraged to work and improve upon their communication skills, through proper listening, literature survey and expression through word and print.

Semester-1

Syllabus for Theory Paper

CHM-DSC-1-MAJ/MIN-101
General Chemistry-1

End semester = 67	**Credits - 03**
Internal Marks = 08	**Contact Hours – 3/week**
Examination Time allowed	**3 Hours**

UNIT I: INORGANIC CHEMISTRY (15 Hrs.)

Atomic Structure (8 Hrs.)

Idea of de-Broglie matter waves, Heisenberg uncertainty principle, atomic orbitals, Schrodinger wave equation, the significance of ψ and ψ^2, quantum numbers, radial and angular wave functions (excluding mathematical relations), probability distribution curves, shapes of *s*, *p*, and *d*orbitals. Aufbau and Pauli exclusion principles, Hund's multiplicity rule, Electronic configurations of the elements and ions.

Periodic Properties (7 Hrs.)

Position of elements in the periodic table, effective nuclear charge and its calculations, Atomic and ionic radii, ionisation energy, electron affinity and electronegativity– definition, methods of determination or evaluation, trends in the periodic table and applications in predicting and explaining the chemical behaviour.

UNIT II: ORGANICCHEMISTRY(15 Hrs.)

Structure and Bonding (5 Hrs.)

Hybridisation, Bond lengths and bond Angles, Bond Energy, Localised and Delocalized chemical bond, Van der Waal's interaction, Resonance and Resonance effect, Hyperconjugation, Inductive and Field effect, Electromeric effect, Hydrogen Bonding.

Reactive Intermediates (5 Hrs.)

Curved arrow notation, Drawing electron movements with arrows, half-headed and double-headed arrows, homolytic and heterolytic bond breaking, Types of Reagents – Electrophiles and nucleophiles, Types of Organic Reactions.Reactive intermediates-Carbocations, Carbanions, Free Radicals, Carbenes, Arynes and Nitrenes (with examples). Assigning Formal charges on intermediates and other ionic species

Geometrical and Conformational Isomerism (5 Hrs.)

Geometrical isomerism

Cause and conditions for geometrical isomerism, Nomenclature of geometrical isomers- cis and trans, E and Z system, Determination of configuration of geometrical isomers.

Conformational Isomerism

Representation of conformations-Sawhorse and Newman formulae, Conformational analysis of ethane, propane, n-butane, cyclohexane, Equatorial and Axial bonds.Conformations of Mono and di-substituted derivatives of cyclohexanes.

UNIT-III PHYSICAL CHEMISTRY (15 Hrs.)

Basic Concepts of Mathematics

(3Hrs.)

Logarithmic relations, differentiation and integration of functions like e^x, x^n, sin x, and log x. Terms of mean and median, precision and accuracy in chemical analysis.

Chemical Kinetics-I(6 Hrs.)

Chemical kinetics and its scope, rate of a reaction, factors influencing the rate of a reaction- concentration,temperature, pressure, solvent, light, and catalyst. Concentration dependence of rates,mathematical characteristicsof simple chemical reactions – zero order, first order, second order, pseudo order, half-life, and mean life.Determination of the order of reaction – differential method, method of integration,method of half-life periodandisolation method, Radioactivedecayasafirst-orderphenomenon.

Chemical Kinetics-II(6 Hrs.)

Theories of Chemical Kinetics: Effect of temperature on rate of reaction, Arrhenius equation, concept of activation energy, Simple collision theory based on hard sphere model, transition state theory (equilibrium hypothesis), Expression for the rate constant based on equilibrium constant and thermodynamic aspects.

Catalysis and general characteristics of catalytic reactions,Homogeneous catalysis, acid-base catalysis and enzyme catalysis,includingtheirmechanisms,Michaelis-Mentenequationforenzymecatalysisanditsmechanism.

Books/Literature suggested

1. Cotton,F.A.,Wilkinson,G.,Gaus,P.L.,BasicInorganicChemistry;2nd edition,Pubs:JohnWileyand Sons, 1995.
2. Lee,J.D.,ConciseInorganicChemistry;4th edition,Pubs:Chapman&HallLtd., 1991.
3. Shriver,D.E.,Atkins,P.W.,InorganicChemistry;4th edition,Pubs:OxfordUniversityPress,2006.
4. Douglas, B., Medaniel, D., Atenander, J., Concepts and Models of Inorganic Chemistry; 3rd edition, Pubs: John Wiley and Sons Inc., 1994,
5. Porterfield,W.W.,Wesky,A.,InorganicChemistry;Pubs:Addison-WesleyPublishingCompany,1984.
6. Miessler,G.L.,Tarr,D.A.,InorganicChemistry;3rd edition,Pubs:PearsonEducationInc.,2004.
7. Puri,B.R.,Sharma,L.R.,Kalia,K.K.,PrinciplesofInorganicChemistry;30th edition,Pubs:Milestones Publisher, 2006-07
8. Atkins, P., Paula, J.de, Atkins Physical Chemistry; 8th edition, Pubs: Oxford University Press, 2008.
9. Puri, B.R., Sharma, L.R., Pathania, M.S., Principles of Physical Chemistry; 43rd edition, Pubs: Vishal Publishing Co., 2008.
10. Barrow, G.M., Physical Chemistry; 6th edition, Pubs: McGraw-Hill Inc, 1996.
11. Berry, R.S., Rice, S.A., Ross, J., Physical Chemistry; 2nd edition, Pubs: Oxford University Press, 2000.
12. Albert, R.A., Silbey, R.J., Physical Chemistry; 1st edition, Pubs: John Wiley & Sons Inc., 1992.
13. Dogra, S.K., Dogra, S., Physical Chemistry Through Problems; Pubs:Wiley Eastern Limited, 1991.
14. Levine, I.N., Physical Chemistry; 5th edition, Pubs: Tata McGraw-Hill Publishing Co. Ltd., 2002.
15. Moore, W. J., Basic Physical Chemistry; Pubs: Prentice Hall of India Pvt. Ltd, 1983.
16. Metz, C.R., Theory and Problems of Physical Chemistry; Schaum's outline series, 2nd edition, Pubs: McGraw-Hall Book Company, 1989.
17. Morrison,R.T.,Boyd,R.N.,OrganicChemistry;6th edition,Pubs:Prentice-Hall,1992.
18. Solomons,T.W.,Fryhle,C.B.,OrganicChemistry;9th edition,Pubs:WileyIndia,2007.
19. WadeJr.,L.G.,Singh,M.S.,OrganicChemistry;6th edition,Pubs:PearsonEducation,2008.
20. Mukherji,S.M.,Singh,S.P.,Kapoor,R.P.,OrganicChemistry;Pubs:NewAgeInternational,1985.
21. Carey,F.A.,Sundberg,R.J.,AdvancedOrganicChemistryPartB:Reactions andSynthesis;5th edition, Pubs: Springer, 2007

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit**(of **19 marks**) will consist of two types of questions:**9 multiple-choice questions**of one mark each and**5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:**the compulsory question and one question each from the UNIT-1, UNIT-2 and UNIT-3.
2. The use of scientific calculators is not allowed.

Semester-1
Syllabus for Practical Paper
CHM-DSC-1-MAJ/MIN-101
General Chemistry-1

End semester = 22	**Credit - 01**
Internal Marks = 03	**Contact Hours – 2/week**
Examination Time allowed	**3 Hours**

I. INORGANIC CHEMISTRY

1. Preparation of standard solutions.
2. Acid-base titrations
    a. HCl vs NaOH
    b. Na_2CO_3 vs HCl
    c. Mixture of bases versus HCl

II. ORGANIC CHEMISTRY

3. Calibration of the thermometer: 80-82° (Naphthalene), 113-114°C(acetanilide),132.5-133°C(Urea), 100°C (distilled Water)
4. Determination of melting pointsof some organic compounds: Naphthalene 80-82°, Benzoic acid 121.5-122° Urea, 132.5-133°, Cinnamic acid 132.5-133°, Acetanilide 113-5-114°, p-Dichlorobenzene 52°. Aspirin135^o.

III. PHYSICAL CHEMISTRY

5. Viscosity: To determine the viscosity of Brine Solution(20%), n-Butylalcohol, cyclohexane

Books/Literature suggested

1. Jolly,W.L., ModernInorganicChemistry;2ndedition,Pubs:TataMcGraw-HillPublishingCompany Limited, 1991.
2. Practical Organic Chemistry, 4TH EDN, F G Mann & B C Saunders,Cambridge. March 1973
3. Vogel's Textbook of Practical Organic Chemistry, 5th Ed., Brian S Furniss, Anthony J Hannaford, Peter W G Smith & Austin R Tatchell,
4. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11th edition, Pubs: R.Chand& Co., New Delhi, 2002.
5. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw-Hill Publishing Co. Ltd., 1983.
6. Levitt, B.P., Findlays Practical Physical Chemistry; 8th edition, Pubs: Longman Group Ltd., London & New York, 1978.

Semester	**II**
Code of Paper	**CHM-DSC-2-MAJ/MIN-201**
Title of Paper	**General Chemistry-2**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100(75Theory + 25 Practical)**
Marks distribution of the Theory paper	**End semester = 67** **Internal Marks = 08**
Marks distribution of the Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. per week** **Practical: 2 Hrs. per week**

Applicable to the students

1. **Those who have opted for chemistry as a Major subject in Table 1 or Table 2**
2. **Those who have opted for chemistry asa Minor subject in Table 1 or Table 2**

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

By the end of the course, students will be able to:

- Explain various theories in covalent-bonded simple inorganic compounds.
- Explain the different concepts regarding structures in ionic solids, their rules and limitations. Lattice energy, Born-Haber cycle, Fajan's rule and Van der Waals forces.
- Understand and explain various methods for preparation, chemical reactions of alkanes and cycloalkanes. Concept of reactivity/selectivity of alkane halogenation & strain theories of cycloalkanes.
- Explain the factors that affect optical isomerism in organic compounds.
- Know the fundamentals of the kinetic theory of gases, Maxwell-Boltzmann distribution law, mean free path of molecular collisions, viscosity, thermal conductivity and diffusion.
- Demonstrate gas laws and isotherms of Van der Waal's, get an essence of the basic concepts of thermodynamics, the first law of thermodynamics, the concept of entropy and the associated theorems, the thermodynamic potentials and their physical interpretations along with Joule-Thomson coefficient and inversion temperature.
- Experimentally the students will be able to explain the concept of standardisation of primary standards, determination of strength & water of crystallisation.

GENERIC OUTCOMES

The first year of the course is designedto inculcate a scientific temper amongst the students. This is done by teaching them how to collect samples/data/evidence, carry out analysis, evaluation and draw logical conclusions.To hone their critical thinking and enhance creativity, which shall lead to innovative and out-of-the-box thinking, which shall help in complex problem-solving situations in higher classes.They are also encouraged to work and improve upon their communication skills, through proper listening, literature survey and expression through word and print.

Semester-2
Syllabus for Theory Paper
CHM-DSC-2-MAJ/MIN-201
General Chemistry-2

End semester = 67	**Credits - 03**
Internal Marks = 08	**Contact Hours – 3/week**
Examination Time allowed	**3 Hours**

UNIT I: INORGANIC CHEMISTRY (15 Hrs.)

Chemical Bonding – I (8 Hrs.)

Covalent Bond – Valence bond theory and its limitations, directional characteristics of covalent bond, various types of hybridisation and shapes of simple inorganic molecules and covalent ions. BeF_2, BF_3, CH_4, PF_5, SF_6, IF_7, $SnCl_2$, XeF_4, valence shell electron pair repulsion (VSEPR) theory to H_3O^+, SF_4, ClF_3, ICl^- and H_2O. MO theory, homonuclear (elements and ions of $1^{st}$and 2nd row), and heteronuclear(CN, CO^+, CO, CN^-), diatomic molecules, Percentageioniccharacterfromdipolemomentandelectronegativity.

Chemical Bonding – II (7 Hrs.)

Ionic Solids – Concept of close packing, Ionic structures (NaCl type, and Zinc blende),radius ratio rule and coordination number, limitation of radius ratio rule, lattice defects, semiconductors.

Lattice energy and Born-Haber cycle, solvation energy and solubility of ionic solids, polarising power and polarisability of ions, Fajan'srule.Metallic bond-free electron, valence bond and band theories.WeakInteractions – Hydrogen bonding, Van der Waals forces.

UNIT II: ORGANIC CHEMISTRY (15 Hrs.)

Alkanes and Cycloalkanes (8 Hrs.)

Structure, IUPAC Nomenclature and Isomerism, Methods of Preparation (Sabatier and Senderen's reaction, from alkyl halide, Wurtz reaction, Corey House synthesis, Kolbe's method, decarboxylation), Physical and Chemical Properties, Mechanism of Halogenation of Alkanes- Relative reactivities of halogens, alkanes, and different classes of Hydrogen atom; Orientation, Reactivity and Selectivity.

Cycloalkanes, nomenclature, methods of formation, chemical reactions, Baeyer's strain theory and its limitation, ring strain in small rings (cyclopropane and cyclobutane), theory of strainless rings, the case of cyclopropane ring: banana bonds.

Stereochemistry and Optical Isomerism (7 Hrs.)

Concept of Isomerism, Types of isomerism,Tetrahedral concept of carbon, Polarimetry, specific rotation. Optical isomerism- Elements of Symmetry, Molecular Chirality, Enantiomerism, stereogeniccentre, optical activity, chiral and achiral molecules with two stereogeniccentres, Diastereoisomers, Erythro and Threo isomers, Meso compounds. Representation of Stereoisomers- Saw-horse, Newman, Flying edge and Fischer Projection. Number of stereoisomers in chiral compound, Racemisation, Internal and External compensation, Resolution of racemic mixture, Enantioselective synthesis, Walden inversion, Relative and absolute configuration, sequence rules, D & L and R & S system of nomenclature (with examples)

UNIT II: PHYSICAL CHEMISTRY (15 Hrs.)

Gaseous States (8 Hrs.)

Gas laws, Idea and real gases, postulates of kinetic theory of gases, deviation from ideal behavior, Van der Waal's equation of state, Critical Phenomena:PV isotherms of real gases, the isotherms of Van der Waal's equation, relationship between critical constants and Van der Waal's constants, the law of corresponding states, reduced equation of state. MolecularVelocities: Root mean square, average and most probable velocities. Collision number, mean free path and collision diameter, Liquification of gases (based on Joule-Thomson effect).

Thermodynamics-I (7 Hrs.)

Definition of Thermodynamic Terms: System, surroundings, etc. Types of systems, intensive and extensive properties, State and path functions and their differentials. Thermodynamic process.Concept of heat and work.First Law of Thermodynamics: Statement, definition of internal energy and enthalpy, Heat capacity, heat capacities at constant volume and pressure and their relationship, Joule's Law-Joule-Thomson coefficient and inversion temperature. Calculations of w, q, dU and dH for the expansion of ideal gases under isothermal and adiabatic conditions for a reversible process.

Books/Literature suggested

1. Cotton,F.A.,Wilkinson,G.,Gaus,P.L.,BasicInorganicChemistry;2nd edition,Pubs:JohnWileyand Sons, 1995.
2. Lee,J.D.,ConciseInorganicChemistry;4th edition,Pubs:Chapman&HallLtd., 1991.
3. Shriver,D.E.,Atkins,P.W.,InorganicChemistry;4th edition,Pubs:OxfordUniversityPress,2006.
4. Douglas, B., Medaniel, D., Atenander, J., Concepts and Models of Inorganic Chemistry; 3rd edition, Pubs: John Wiley and Sons Inc., 1994,
5. Porterfield,W.W.,Wesky,A.,InorganicChemistry;Pubs:Addison-WesleyPublishingCompany,1984.
6. Miessler,G.L.,Tarr,D.A.,InorganicChemistry;3rd edition,Pubs:PearsonEducationInc.,2004.

7. Puri,B.R.,Sharma,L.R.,Kalia,K.K.,PrinciplesofInorganicChemistry;30th edition,Pubs:Milestones Publisher, 2006-07
8. Atkins, P., Paula, J.de, Atkins Physical Chemistry; 8th edition, Pubs: Oxford University Press, 2008.
9. Puri, B.R., Sharma, L.R., Pathania, M.S., Principles of Physical Chemistry; 43rd edition, Pubs: Vishal Publishing Co., 2008.
10. Barrow, G.M., Physical Chemistry; 6th edition, Pubs: McGraw-Hill Inc, 1996.
11. Berry, R.S., Rice, S.A., Ross, J., Physical Chemistry; 2nd edition, Pubs: Oxford University Press, 2000.
12. Albert, R.A., Silbey, R.J., Physical Chemistry; 1st edition, Pubs: John Wiley & Sons Inc., 1992.
13. Dogra, S.K., Dogra, S., Physical Chemistry Through Problems; Pubs:Wiley Eastern Limited, 1991.
14. Levine, I.N., Physical Chemistry; 5th edition, Pubs: Tata McGraw-Hill Publishing Co. Ltd., 2002.
15. Moore, W. J., Basic Physical Chemistry; Pubs: Prentice Hall of India Pvt. Ltd, 1983.
16. Metz, C.R., Theory and Problems of Physical Chemistry; Schaum's outline series, 2nd edition, Pubs: McGraw-Hall Book Company, 1989.
17. Morrison, R.T., Boyd, R.N., Organic Chemistry; 6th edition, Pubs: Prentice-Hall, 1992.
18. Solomons, T.W., Fryhle, C.B., Organic Chemistry; 9th edition, Pubs: Wiley India, 2007.
19. Wade Jr., L.G., Singh, M.S., Organic Chemistry; 6th edition, Pubs: Pearson education, 2008.
20. Mukherji, S.M., Singh, S.P., Kapoor, R.P., Organic Chemistry; Pubs: New Age International, 1985.
21. Carey, F.A., Sundberg, R.J., Advanced Organic Chemistry Part B: Reactions and Synthesis; 5th edition, Pubs: Springer, 2007

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. The **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **19 marks**) will consist of two types of questions:**9 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:**the**compulsory question** and one question each from the **UNIT-1, UNIT-2 and UNIT-3**.
2. The use of scientific calculators is not allowed.

Semester-2
Syllabus for Practical Paper
CHM-DSC-2-MAJ/MIN-201
General Chemistry-2

End semester = 22	**Credit - 01**
Internal Marks = 03	**Contact Hours – 2/week**
Examination Time allowed	**3 Hours**

I. INORGANIC CHEMISTRY

1. Permanganometry and dichromatometry
   a. Standardization of $KMnO_4$ solution
   b. Determination of the strength of potassium oxalate-oxalic acid mixture/Mohr's salt/ferrous sulphatesolution by usinga standard solution of $KMnO_4$ solution by redox titration.
   c. Determination of water of crystallisation in Mohr's salt and ferrous sulphateby usinga standard solution of $KMnO_4$ by redox titration.

II. ORGANIC CHEMISTRY

1. Concept of induction of crystallisation.
2. Phthalic acid from hot water (using fluted filter paper and a stemless funnel).
3. Acetanilide from boiling water.
4. Benzoic acid from water

III. PHYSICAL CHEMISTRY

Surface Tension: To determine the surface tension of Brine Solution (20%), n-Butyl alcohol, cyclohexane

Books/Literature suggested

1. Jolly,W.L., ModernInorganicChemistry;2nd edition,Pubs:TataMcGraw-HillPublishingCompany Limited, 1991.
2. Practical Organic Chemistry, 4TH EDN, F G Mann & B C Saunders,Cambridge. March 1973
3. Vogel's Textbook of Practical Organic Chemistry, 5th Ed., Brian S Furniss, Anthony J Hannaford, Peter W G Smith & Austin R Tatchell,
4. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11th edition, Pubs: R.Chand& Co., New Delhi, 2002.
5. Das, R.C., Behra, B., Experimental Physical Chemistry; Pubs: Tata McGraw-Hill Publishing Co. Ltd., 1983.
6. Levitt, B.P., Findlay's Practical Physical Chemistry; 8th edition, Pubs: Longman Group Ltd., London & New York, 1978.

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COMPUTER SCIENCE

NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE/ DOUBLE MINOR (Table 1*)
and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY (Table 2*)
BSc Computer Science subjects; For session 2026-27

Table 1*						
Sem	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr)**		**Total Credits**
I	L-100	CSA-DSC-1(Maj/Min)-101	Computer Fundamentals and PC Software	2+2	4(x3)	24
		CSA-MDC-101/201/301	E-Commerce	3+0	3	
		CSA-MDC-102/202/302	Management Information System	3+0		
		CSA-MDC-103/203/303	Web Development/ Designing	0+3		
		CSA-MDC-104/204/304	Introduction to Information Technology	0+3		
		CSA-MDC-105/205/305	Computer Programming and Problem Solving	3+0		
		CSA-MDC-106/206/306	Data Analysis using Excel	0+3		
		CSA-VAC-101/201/301/401	Introduction to Blockchain Technology	2+0	2	
		CSA-VAC-102/202/302/402	Introduction to Computer Hardware	2+0		
		CSA-VAC-103/203/303/403	Digital Empowerment	2+0		
		CSA-VAC-104/204/304/404	Ethics of Artificial Intelligence	2+0		
		CSA-SEC-101/201/301	Computer Fundamentals and Personal Computing Software	0+3	3	
		CSA-SEC-	Desktop	0+3		

Table 2*						
Sem	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**		**Total Credits**
I	L-100	CSA-DSC-1(Maj/Min)-101	Computer Fundamentals and PC Software	2+2	4(x3)	24
		CSA-MDC-101/201/301	E-Commerce	3+0	3	
		CSA-MDC-102/202/302	Management Information System	3+0		
		CSA-MDC-103/203/303	Web Development/ Designing	0+3		
		CSA-MDC-104/204/304	Introduction to Information Technology	0+3		
		CSA-MDC-105/205/305	Computer Programming and Problem Solving	3+0		
		CSA-MDC-106/206/306	Data Analysis using Excel	0+3		
		CSA-VAC-101/201/301/401	Introduction to Blockchain Technology	2+0	2	
		CSA-VAC-102/202/302/402	Introduction to Computer Hardware	2+0		
		CSA-VAC-103/203/303/403	Digital Empowerment	2+0		
		CSA-VAC-104/204/304/404	Ethics of Artificial Intelligence	2+0		
		CSA-SEC-101/201/301	Computer Fundamentals and Personal Computing Software	0+3	3	
		CSA-SEC-102/202/302	Desktop Publishing using GIMP	0+3		
		CSA-SEC-103/203/303	Backend Web Development	0+3		
		CSA-SEC-	Business	0+3		

		102/202/302	Publishing using GIMP						104/204/304	Intelligence and Data Visualization				
		CSA-SEC-103/203/303	Backend Web Development	0+3					CSA-SEC-105/205/305	Digital Marketing	0+3			
		CSA-SEC-104/204/304	Business Intelligence and Data Visualization	0+3					CSA-SEC-106/206/306	Computer Literacy	0+3			
									CSA-SEC-107/207/307	Desktop Publishing Using Photoshop	0+3			
									CSA-SEC-108/208/308	Fundamentals of Web Designing	0+3			
		CSA-SEC-105/205/305	Digital Marketing	0+3					CSA-SEC-109/209/309	Application Software for Office Automation	0+3			
		CSA-SEC-106/206/306	Computer Literacy	0+3					CSA-SEC-110/210/310	Python Programming	0+3			
		CSA-SEC-107/207/307	Desktop Publishing Using Photoshop	0+3										
		CSA-SEC-108/208/308	Fundamentals of Web Designing	0+3										
		CSA-SEC-109/209/309	Application Software for Office Automation	0+3										
		CSA-SEC-110/210/310	Python Programming	0+3										
		(AEC-1-101)	Eng-1	2					(AEC-1-101)	Eng-1	2			
		(PUN-COMPL-1-101/HCP-COMPL-1-101)	Pbi-1/HCP-1	2					(PUN-COMPL-1-101/HCP-COMPL-1-101)	Pbi-1/HCP-1	2			
II	L-100	CSA-DSC-2(Maj/Min)-201	Programming in C	2+2	4 (x3)	24	II	L-100	CSA-DSC-2(Maj/Min)-201	Programming in C	2+2	4 (x3)	24	
		CSA-MDC-101/201/301	E-Commerce	3+0	3				CSA-MDC-101/201/301	E-Commerce	3+0	3		
		CSA-MDC-102/202/302	Management Information System	3+0					CSA-MDC-102/202/302	Management Information System	3+0			
		CSA-MDC-103/203/303	Web Development/ Designing	0+3					CSA-MDC-103/203/303	Web Development/ Designing	0+3			
		CSA-MDC-104/204/304	Introduction to Information Technology	0+3					CSA-MDC-104/204/304	Introduction to Information Technology	0+3			
		CSA-MDC-105/205/305	Computer Programming and Problem Solving	3+0					CSA-MDC-105/205/305	Computer Programming and Problem Solving	3+0			
		CSA-MDC-106/206/306	Data Analysis using Excel	0+3					CSA-MDC-106/206/306	Data Analysis using Excel	0+3			
		CSA-VAC-101/201/301/401	Introduction to Blockchain Technology	2+0	2				CSA-VAC-101/201/301/401	Introduction to Blockchain Technology	2+0	2		

		CSA-VAC-102/202/302/402	Introduction to Computer Hardware	2+0						CSA-VAC-102/202/302/402	Introduction to Computer Hardware	2+0		
		CSA-VAC-103/203/303/403	Digital Empowerment	2+0						CSA-VAC-103/203/303/403	Digital Empowerment	2+0		
		CSA-VAC-104/204/304/404	Ethics of Artificial Intelligence	2+0						CSA-VAC-104/204/304/404	Ethics of Artificial Intelligence	2+0		
		CSA-SEC-1101/201/301	Computer Fundamentals and Personal Computing Software	0+3	3					CSA-SEC-101/201/301	Computer Fundamentals and Personal Computing Software	0+3	3	
		CSA-SEC-102/202/302	Desktop Publishing using GIMP	0+3						CSA-SEC-102/202/302	Desktop Publishing using GIMP	0+3		
		CSA-SEC-103/203/303	Backend Web Development	0+3						CSA-SEC-103/203/303	Backend Web Development	0+3		
		CSA-SEC-104/204/304	Business Intelligence and Data Visualization	0+3						CSA-SEC-104/204/304	Business Intelligence and Data Visualization	0+3		
		CSA-SEC-105/205/305	Digital Marketing	0+3						CSA-SEC-105/205/305	Digital Marketing	0+3		
		CSA-SEC-106/206/306	Computer Literacy	0+3						CSA-SEC-106/206/306	Computer Literacy	0+3		
		CSA-SEC-107/207/307	Desktop Publishing Using Photoshop	0+3						CSA-SEC-107/207/307	Desktop Publishing Using Photoshop	0+3		
		CSA-SEC-108/208/308	Fundamentals of Web Designing	0+3						CSA-SEC-108/208/308	Fundamentals of Web Designing	0+3		
		CSA-SEC-109/209/309	Application Software for Office Automation	0+3						CSA-SEC-109/209/309	Application Software for Office Automation	0+3		
		CSA-SEC-110/210/310	Python Programming	0+3						CSA-SEC-110/210/310	Python Programming	0+3		
		(AEC-2-201)	Eng-2	2						(AEC-2-201)	Eng-2	2		
		(PUN-COMPL-2-201/HCP-COMPL-2-201)	Pbi-2/HCP-2	2						(PUN-COMPL-2-201/HCP-COMPL-2-201)	Pbi-2/HCP-2	2		
The candidate has to select Major and Minor out of 3 subjects studied in the I and II Semester at the end of the Semester II														
The studentexitingtheprogramme after oneyear, securing48credits,willbeawardedUG Certificateprovidedhe/shesecures 4creditsfromsummer internship/community work.														

***Note: Student need to select one MDC, VAC, SEC out of given options, respectively. NO MDC, SEC, VAC will be repeated. All MDC, SEC and VAC are Level-100**

***Note: The candidate will opt for either Template 1 or 2 from Semester III onwards.**

Notes:

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

Computer Science and Applications

Paper Title:ComputerFundamentalsandPCSoftware
Paper Code: CSA-DSC-1(Maj/Min)-101

Th	Pr	Cr	
2	2	4	TheoryExternalMarks:45
			TheoryInternalMarks:5
Level: L- 100			NumberofTheoryHours:30
			Practical External Marks: 50
			Practical External hours: 60

Time Duration:3Hrs.
Objectives:The objective of this course is to familiarize students with the fundamentals of computers and packages commonly used in application software.

Learning Outcome:
On successful completion of the course, students will be able to:

- Understand the fundamentals of personal computers, application usages etc.
- Get the knowledge of peripherals of personal computer and booting process.
- Useword processing packages for writing mails,lettersand chapters etc.
- Handle statistical data using spreadsheets packages.
- Design presentations for business analyses, industrial growth etc.

Note:

- TheQuestionPaperwillconsistofFour Units.
- ExaminerwillsettotalofNINEquestionscomprisingTWOquestionsfromeachUnitandONE compulsory question of short answer type covering whole syllabi.
- Thestudentsarerequiredtoattempt ONEquestionfromeachUnitandtheCompulsoryquestion.
- Allquestionscarryequalmarksunlessspecified.

UNIT-I

Introduction to Computers: Characteristics of Computers, Functional Units of Computer System: CPU, registers, systembus, mainmemory unit, cachememory; Insideacomputer: SMPS, Motherboard, Portsand Interfaces, expansion cards, ribbon cables, memory chips, processors

Devices:Inputand output devices, Keyboard, Mouse, Joystick, Scanner, OCR, OMR, BarCode Reader, Web Camera, Monitor, Printer and Plotter; Memory: Primary, Secondary/Auxiliary memory, RAM and ROM and their types, Cache memory, Units of measurement of storage; Storage devices, Hard disks, Optical disks, Memory storage devices: Flash Drive, Memory card; Types of software: System and Application software

DataRepresentation:Bit,Byte,Binary,Decimal,Hexadecimal,andOctalSystems.Conversionsbetween different number systems; Character Codes: Introduction to ASCII, EBCDIC and Unicode character sets

UnderstandingWindowsOperatingSystem:Anatomyofwindows,TaskbarSettings,managingfoldersand files, Searching Files and folders, Customizing Windows, Recycle Bin, Control Panel.

UNIT-II

IntroductiontoWordProcessing:Opening,Creating,Saving,PrintingandclosingDocumen ts,Usingthe Interface (Menu Toolbars), Editing Text (Copy, Delete, Move), Finding and Replacing Text, Spell Check, Autocorrect; Auto-text, Character formatting, Paragraph formatting, Page formatting.

DocumentEnhancement:InsertingPictures,Tables,insertingspecialcharacters,AddingBord ersandshading, Adding Headers and Footers, numbering pages, setting up Multiple columns, Adjusting Margins and Hyphenating Documents, changing document views, Adding comments to a document, Tracking changes to a document.

Usingstylesandtemplates:Applying,modifyingand creatingnew(custom)styles;Creatingatableof contents, creating indexes and bibliographies

MailMergeandMacros:CreatingMasterDocumentandDataSource,Mergingandprinting Documents; Creating and using Macros, Creating fill-in forms.

UNIT-III

IntroductiontoSpreadsheets: Worksheetoverview,Row,Column,Cells,Ribbon,Tabs,Groups,NameBoxand Formula Bar; Creating, Opening, Saving and printing worksheet; Speeding up data entry: using fill tool, fillseries,definingfillseries;workingwithFormulae,Dataformatting(numberformatting,datef ormatting),Working with Ranges, Absolute, relative and Mixed addressing; Viewing a spreadsheet: freezing rows and columns, splitting screen; creating, sorting and filtering Data Base; Creating and running Macros, Assigning shortcuts to Macros

UsingFunctions:Statistical,financial,Mathematical,string,dateandtime.

Workingwithsheets:insertingnewsheet,moving,deletingandrenamingsheets

CreatingCharts:UnderstandingCharttypes,CreatingEmbeddedCharts,formattingCharts:C hangingChart types, adding Titles, Legends and Gridlines, Printing Charts

UNIT–IV

PresentationSoftware:Creating,saving,andprintingpresentations;selectingdesigntemplates,C hanging Background and Layout, Inserting tables and images, animations and transitions Databases: Introduction to database, Creating database using Wizard or from scratch, creating tables using wizard,enteringdata,usingdesignview,saving,inserting,editing,Changingpropertiesoffield s,settingprimary key.

SuggestedReadings:

1. SinhaP.K.&SinhaPriti,ComputerFundamentals,BPBPublications
2. Prof.SatishJain,ShashiSingh,M.Geetha,Windows10withMSoffice2016,BPBPublications
3. MansfieldRon,WorkinginMSOffice,TataMcGrawHill.

Paper Code: CSA-DSC-1(Maj/Min)-101P
(Practical based on paper CSA-DSC-1(Maj/Min)-101)

TimeDuration:3Hrs.

Note: Paper will be set at the time of examination. Dueweight-agemay be given to the practical note-book and Assignments in evaluation.

Paper Title:Programming in C
Paper Code:CSA-DSC-2(Maj/Min)-201

Th	Pr	Cr	TheoryExternalMarks:45
2	2	4	TheoryInternalMarks:5
Level: L- 100			NumberofTheoryHours:30
			Practical External Marks: 50
			Practical External hours: 60

TimeDuration:3Hrs.

Objectives: The objective of this course is to familiarize students with programming language ”C”anditsusein Problem solving.

Learning Outcome:

On successful completion of the course, students will be able to:

- Understand the programming concepts.
- Learn the C programming in practical usage.
- Solve programming problems using C programming language.
- Gain the in depth knowledge of strings,arrays,functions and file handling etc.
- Develop applications based on programming concepts.

Note:

- The Question Paper will consist of Four Units.
- Examiner will set total of NINE questions comprising TWO questions from each Unit and ONE compulsory question of short answer type covering whole syllabi.
- The students are required to attempt ONE question from each Unit and the Compulsory question.
- All questions carry equal marks unless specified.

UNIT-I

Programming Process: Steps indeveloping of a program,Algorithm development, Flowchart, Testing and Debugging.

Fundamentals of C Language: History of C, Character Set, Identifiers and Keywords, Constants,Types of C Constants, Rules for Constructing Integer, Real and character Constants, Variables, Data Types, rules for constructing variables.

Operators and Expressions: C Instructions, Arithmetic operators, Relational operators, Logical operators, Assignment Operators, Type Conversionin Assignments, Hierarchy of Operations, Standardand Formatted Statements, Structure of a C program, Compilation and Execution.

UNIT-II

Decision Control Structure: Decision making with IF- statement, IF-Else and Nested IF-Else, The elseif Clause. Loop Control Structure: While and do-while, for loop and Nested for loop, Case Control Structure: Decision using switch, The goto statement. Functions:Libraryfunctionsanduserdefinedfunctions,GlobalandLocalvariables,Function Declaration, Calling and definition of function, Methods of parameter passing to functions, recursion.

UNIT-III

Arrays: Introduction, Array declaration, accessing values in an array, initializing values in an array, Single- and Two-Dimensional Arrays, initializing a2-Dimensional Array, Memory Mapofa2-Dimensional Array, Arrays of characters, Insertion and deletion operations, Searching the elements in an array, Using matrices in arrays.
Pointers:Pointerdeclaration,Addressoperator"&",Indirectionoperator"*",ArrayofPointers.

UNIT-IV

StorageClassesinC:Auto,Extern,RegisterandStaticwiththeirScope,Storage&Lifetime. StringManipulationinC:DeclaringandInitializingstringvariables,Readingandwritingstris, String Handling functions (strlen(), strcpy(), strcmp(), strcat()). Structures and Unions: Definition. Advantages of Structure, Declaration of structures, Structure Initialization, Accessing Structure members, Arrays of Structures, Union Definition, Difference between Structureand Union.

SuggestedReadings:

1. YashavantP.Kanetkar:LetusC,BPBPublications,NewDelhi.
2. Salaria,R.S.:TestYourSkillsinC,SalariaPublications,New Delhi.
3. C.Balaguruswami:ProgrammingwithCLanguage,TataMcGrawHill,NewDelhi.
4. ByronS.Gottfried:ProgramminginC,McGrawHillsPublishers,NewYork.
5. M.T.Somashekara:ProgramminginC,PrenticeHallof India.

PaperCode:CSA-DSC-2(Maj/Min)-201P
(Practical based on paper CSA-DSC-2(Maj/Min)-201)

TimeDuration:3Hrs.

Note: Paperwillbesetatthetimeofexamination.Dueweight-agemaybegiven tothepracticalnote-bookand Assignments in evaluation.

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ENVIRONMENT CONSERVATION

SEMESTER-I
MINOR COURSE
NATURAL RESOURCE MANAGEMENT

Course Code: MINC1 Credits= 4 (T=3 + P=1)
Total Hours = 60 (Theory=45+Practical=15)
Total Marks = 100 (Theory=75 [Final =60; Internal Assessment =15]+Practical=25)

Course Outcomes

1. *The students will be able to understand about natural resources, their types and realize the importance of resources.*
2. *The students will be able to acquire deep knowledge about land and forest resources.*
3. *The students will be able to describe energy resources.*
4. *The students will be able to analyse Indian as well as global statusof natural resources.*
5. *The students will be able to study conservation methods of natural resources.*

THEORY

Unit-I (15 Hrs)
Energy Resources:
1.1 Natural resources-Classification and types of resources
1.2 Renewable and non-renewable resources
1.3 Concept of conventional and non-conventional sources of energy
1.4 Fossil fuels- coal, crude oils and natural gas
1.5 Solar energy
1.6 Wind energy
1.7 Biomass energy
1.8 Geothermal energy

Unit-II (15 Hrs)
Forest Resources
2.1 Importance of forest;
2.2 Productive functions, protective and ameliorative functions,
2.3 Non-timber forest products
2.4 Major forest types in India
2.5 Conservation of forest resources
2.6 Global status of forest conservation efforts

Unit-III (15 Hrs)
Land Resources
3.1 Land Resources: Global land use patterns
3.2 Soil degradation – causes; Soil erosion, Desertification
3.3 Wasteland Reclamation. Land degradation
3.4 Land cover, land use changes; land degradation and its control
3.5 Water logging and salinity
3.6 Organic farming

Unit-IV (15 Hrs)
Water Resources
1.1 Composition of Hydrosphere, Global Water Balance
1.2 Hydrological Cycle; Major Sources of Water, Surface and subsurface water sources
1.3 Water Anomalies including Floods and Droughts
1.4 Overutilization of Water and Water Scarcity
1.5 Conflicts Over Water – National and International
1.6 Case Studies; Satluj Yamuna Link Canal Issue, Kaveri Issue

PRACTICALS (15 Hrs)
1. To do a survey of water consumption and conservation practises of a selected area.
2. To visit and make a list of forest and non-forest products in the market.
3. Identify the sources of energy used in your area
4. Visit to a local area to document environmental assets river/ forest/grassland/hill/mountain
5. To determine soil quality parameters using portable soil analysis kit
6. Draw the pie diagram showing the global water balance.

SUGGESTED READINGS:
1. Rogoff, M., Screve, F. 2019. Waste-to-Energy, Elsevier, Amsterdam
2. Singh, M.P, Rallan,B.R. and Vivek Kumar. 2012. Natural Resources Management. Emkay Publishing House.
3. Odum, E.P. and Barrett, G.W. 2008. Fundamentals of Ecology (5thEdition). Thomson Brooks Australia and Affiliated to East West Press Pvt. Ltd., New Delhi.
4. Kanagasabai, T. 2010. Environmental Studies. PHI Learning Private Limited, New Delhi. 5. Sharma, P.D. 2018. Ecology and Environment. Rastogi Publication, Meerut.
6. Joseph, K. and Nagendran, R. 2004. Essentials of Environmental Studies. Pearson Education, New Delhi.

Instructions for paper setter for Theory Paper:
There will be nine questions in all. Question # 1 shall be compulsory comprising of four short answer questionscovering whole of the syllabus. There shall be two questions each from Unit I to IV. The candidate shall be required to attempt compulsory question, and one question each from Unit I to IV. Each question shall be of 12 marks.

SEMESTER-II
MINOR COURSE
ENVIRONMENTAL POLLUTION AND MANAGEMENT

CourseCode:MINC2 Credits= 4 (T=3 + P=1)

Total Hours = 60 (Theory=45+Practical=15)

Total Marks = 100 (Theory=75 [Final =60; Internal Assessment =15] +Practical=25)

Course Outcomes

1. *The students will be able to understand the causes, sources and effects of environmental pollution.*
2. *Thestudentswillbeableto learn about atmospheric structure and processes*
3. *ThestudentswillacquireknowledgeaboutGlobal Water Balance and Hydrological cycle.*
4. *Thestudentswill get practical knowledge of water quality parameters and soil parameters.*

THEORY

Unit-I(15Hrs)
Environmental Pollution

1.1 Concept of Sound and Noise and Measurement
1.2 Noise Pollution; Anthropogenic and Natural Sources
1.3 Effects of Noise Pollution
1.4 Control of Noise pollution
1.5 Radiation Pollution ; Concept of Sources
1.6 Effects of Radiation Pollution and Control Measures

Unit-II(15Hours)
Mineral Extraction and Management

2.1 Concept of Minerals and Ores
2.2 Process of Mineral Extraction
2.3 Mining and Types of Mines
2.4 Impact of Mining on Environment
2.5 Reclamation of Mines

Unit-III(15 Hrs)
Air Pollution and Control Measures

3.1 Composition and Structure of Atmosphere
3.2 Air Pollution: Concept and Sources; Noise Pollution
3.3 Air Pollutants and Categories
3.4 Effects of Air Pollution
3.5 Acid rain
3.7 Control Measures; Air Pollution Control Devices

Unit-IV (15 Hrs)
Water Pollution and Control Measures

4.1 Water Pollution: Concept, Causes and Sources
4.2 Water Pollutants and categories
4.3 Effect of water pollution on plants , animals and humans
4.4 Basics of Water Testing , Potable water
4.5 Control Measures; Wastewater Treatment
4.6 Root Zone technology

PRACTICALS (15 Hrs)

1. To prepare a report on Noise Pollution, its sources and effects in a given area
2. To determine water quality parameters (pH, EC and TDS) using analysis kit.
3. To study air quality parameter like SPM using Air Quality Monitor
4. To visit, collect data and prepare reports regarding air and water pollution of a site.
5. Tostudy and do surveys on major minerals and their ores.

SUGGESTED READINGS:

1. Joseph, K. and Nagendran, R.(2004). Essentials of Environmental Studies. Pearson Education, New Delhi.
2. Cunningham, W.P. and Cunningham, M.A. (2002*). Environmental Science: Inquiry andApplications.* AGlobalConcern.TataMcGraw-HillPublishingCompany,NewDelhi.
3. De. AK, (2003). Environmental Chemistry. New Age International, New Delhi
4. Khitoliya, RK (2012) Textbook on Environmental Pollution. S Chand Publishing, New Delhi
5. American Public Health Association, Standard Methods for Examination of Water and Wastewater, APHA, New York, 2003.
6. Sengupta, B., John, S. & Abraham, T. Forth Estate, A Handbook on Air, Water and Land Pollution. IDMA Foundation for Sustainable Development and BR Publications Corporation, Delhi. 2006.
7. Aggarwal, A. K., Noise Pollution. APH Publications, New Delhi, 2008.
8. Dara, S.S., A Textbook of Environmental Chemistry. 2nd Edition. S. Chand & Company Ltd., Ram Nagar, New Delhi, 2010.
9. Hill, M. K., Understanding Environmental Pollution, Cambridge Univ. Press, 2008.
10. Aggarwal, A. K., Water Pollution, APH Publications, New Delhi, 2008

Instructions for paper setter for Theory Paper:

There will be nine questions in all. Question # 1 shall be compulsory comprising of four short answer questions covering whole of the syllabus. There shall be two questions each from Unit I to IV. The candidate shall be required to attempt compulsory question, and one question each from Unit I to IV. Each question shall be of 12 marks.

INDUSTRIAL CHEMISTRY

CURRICULUM AND CREDIT DISTRIBUTION FOR INDUSTRIAL CHEMISTRY AS **MAJOR** SUBJECT AS PER TABLE 2

Table 2

Sem	Level	Course Code	Course Name	Credits	Total Credits
I	L-100	(ICH-DSC-1 MAJ/MIN-101)	Basic Industrial Principles, Data Analysis and Laboratory Operations	4	24
		B1	To be chosen by the student	4	
		C1	To be chosen by the student	4	
		IDC-1	To be chosen from the available options	3	
		SEC-1	To be chosen from available options	3	
		ENG-1	Compulsory	2	
		VAC-1	To be chosen from the common basket	2	
		PBI-1/HCP-1	Compulsory	2	
II	L-100	(ICH-DSC-2 MAJ/MIN-201)	Applied Industrial Chemistry	4	24
		B2	To be chosen by the student	4	
		C2	To be chosen by the student	4	
		IDC-2	To be chosen from the available options	3	
		SEC-2	To be chosen from available options	3	
		ENG-2	Compulsory	2	
		VAC-2	To be chosen from the common basket	2	
		PBI-2/HCP-2	Compulsory	2	

CURRICULUM AND CREDIT DISTRIBUTION FOR INDUSTRIAL CHEMISTRY AS **MINOR** SUBJECT AS PER TABLE 1 AND TABLE 2

Table 2

Sem	Level	Course Code	Course Name	Credits
I	L-100	(ICH-DSC-1 MIN/MAJ-101)	Basic Industrial Principles, Data Analysis and Laboratory Operations	4
II	L-100	(ICH-DSC-2 MIN/MAJ-201)	Applied Industrial Chemistry	4

Note:-

1. In the first semester, students are offered three different subjects: A, B & C.
2. At the end of the second semester, each student has the option to choose any one of the two streams, namely Single Major or Multi-Disciplinary. The student choosing the Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student choosing the Multi-Disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in the Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for a 4-year degree B.SC HONOURS with research, on the condition that the Institution is a PU-approved Research Centre
5. The student opting for a Single Major must secure 50% of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for a Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year, securing 48 credits, will be awarded a UG Certificate provided he/she secure 4 credits from summer internship/community work. Further, the student exiting the programme after two years, securing 96 credits, will be awarded a UG Diploma provided he/she secure 4 credits from a summer internship. The student successfully securing 144 credits will be awarded a BA/BSc degree.
7. Language papers chosen as MIL-1 and MIL-2 should be on the same subject.
8. Once Major and Minor papers are chosen in the 3rd semester, the same will continue as Major and Minor subjects in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write a Dissertation of 08 credits in one of the two Major subjects.

Semester	**I**
Code of Paper	**ICH-DSC-1-MAJ/MIN-101**
Title of Paper	**Basic Industrial Principles, Data Analysis and Laboratory Operations**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100 (75 Theory + 25 Practical)**
Marks distribution of the Theory paper	**End semester = 67 Internal Marks = 08**
Marks distribution of the Practical paper	**End semester = 22 Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. per week Practical: 3 Hrs. per week**

Applicable to the students

3. **Those who have opted for Industrial chemistry as a Major subject in Table 2.**
4. **Those who have opted for Industrial chemistry as a Minor subject in Table 1 or Table 2.**

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

By studying this course, the students will be able to:

- Explain the what are the basic dimensions and units.
- The student shall be able to carry out the important chemical calculations and determine the errors that may result in the data.
- Explain the theory behind chemical concentrations, preparation of standards and analyse data.
- The student will be able to explain safety of chemical handling through the comprehensive understanding of laboratory equipment and glassware.
- Emphasis is given to developing practical skills that will aid the student to set up accurate and reproducible, result-oriented experiments in higher classes.

GENERIC OUTCOMES

The first year of the course is designed to heighten the chemical temperament amongst the students. This is done by teaching them how to collect samples/data/evidence, carry out data analysis, evaluation and draw logical conclusions. They are also encouraged to work and improve upon their communication skills, through proper listening, plenty of hands-on laboratory exposure, group discussions and individual expression.

For practical work, groups shall be constituted, with student strength not exceeding 20 students per group.

Semester-1
Syllabus for Theory Paper
ICH-DSC-1-MAJ/MIN-101
Basic Industrial Principles, Data Analysis and Laboratory Operations

End semester = 67	**Credits - 03**
Internal Marks = 08	**Contact Hours – 3/week**
Examination Time allowed	**3 Hours**

UNIT-1(15 Hrs.)

Dimensions and Units - Introduction, Definitions of the Basic Units - (Mass, Length, Time, Temperature, Volume, Pressure), Basic terminologies, Characteristics of standard units, Classification of units: Fundamental and Derived units, Systems of units, Basic and supplementary units, Conversion between units and conversion factors, Metric prefixes. Definition of Dimensions, Dimensional Formula, Dimensional Equation, Dimensional Constant, Dimensional Variable, Dimensionless Constant, Dimensionless Variable, Primary Dimensions and units, Secondary Dimensions and units, Dimensional Analysis, Dimensional Homogeneity, Applications of Dimensional Equation, Limitations of Dimensional Equation.

Basic Chemical Calculations - Atomic weight, Molecular weight, Equivalent weight, Mole, Mole fraction, Mole percent, Mass fraction of solution, Weight percentage, Density, Concentration, Specific gravity, Volumetric flow rate, Analyses of Multi Components Solutions and Mixtures.

Errors in Chemical Analysis - Definition of errors, mean and median, Accuracy and Precision in experimental data, Types of errors: Absolute and relative error, Analytical errors: Random, systematic and gross errors, Sources and factors affecting, detection and elimination of errors, significant figures.

UNIT-2(15 Hrs.)

Chemical Concentration – Definition, Concentration Units: Percentage (mass percent, volume percent, Mass-volume percent), Parts per millions (ppm), Strength of solution, Molarity, Molality, Normality, Formality, Factors affecting concentration, Concentration unit conversion and problem-solving in chemical engineering involves conversion between different concentration units like molarity, molality, normality, parts per million (ppm), and mass/volume percentages, Formula weight and molecular weight, Molecular formula and Empirical formulas.

Preparation of solutions -Basic Concepts of preparing solutions, Standard solutions, Primary standards, Secondary standards, Buffer Solutions, Concentrated solutions, Preparation of simple inorganic salt solutions, Preparation of acid solutions, Preparation of Base Solutions.

Stoichiometric Analysis – Calculation of the quantitative relationships between reactants and products in chemical reactions, Reaction stoichiometry, Balanced Chemical Equations for complete and partial combustion of carbon and hydrocarbons, Stoichiometric Coefficients, Stoichiometric Ratios, concept of limiting reactant and excess reactants in a reaction mixture with examples, Yield.

Data analysis and Chemical drawing and visualization tool - To plot linear and non-linear equations using Excel spreadsheet, Analyzing and interpreting experimental data using origin software, Calibration curve plotting using excel and origin software, draw and visualize chemical structures using ChemDraw and ChemSketch.

UNIT-3(15 Hrs.)

Common Laboratory Apparatus and Calibration – Theory of Analytical Balance, Operation and theory of the balance, Construction details, Components, Sources of Errors and Error Analysis, Maintenance Guidelines, identification, proper usage, and calibration techniques of various laboratory tools and equipment.

pH meter - Introduction to pH and pH Measurement, Principle of pH Measurement, pH electrode components, use of pH Meter, Calibration using buffer solutions, Troubleshooting and maintenance of pH meter, application of data Factors affecting pH measurement.

Fundamentals of Laboratory Apparatus - Types of Laboratory apparatus: Volumetric flasks, burettes, pipettes, weighing bottles, specific gravity bottle, different types of funnels, chromatographic columns, chromatographic jars, desiccators, drying ovens, filter crucibles, rubber policeman, use of volumetric glass ware, Importance of laboratory apparatus.

Safety with chemicals and waste - Personal Protective Equipment (PPE), Handling hazardous materials, Emergency Procedures.

Books Recommended

1. Higson, S. P.J. (2003), Analytical Chemistry, Oxford University Press.
2. Skoog, D.A., West, D.M. (2003), Fundamentals of Analytical Chemistry, Brooks/Cole.
3. Christian, G.D. (2004), Analytical Chemistry, 6th Edition, John Wiley & Sons, New York.
4. Fifield, F.W., Kealey, D. (2000), Principles and Practice of Analytical Chemistry, Wiley.
5. Dean, J. A. (1997), Analytical Chemistry Handbook, McGraw Hill.
6. Day, R. A., Underwood, A. L. (1991), Quantitative Analysis, Prentice Hall of India.
7. Gordus, A. A. (1985), Schaum's Outline of Analytical Chemistry, Tala McGraw-Hill.
8. Harris, D. C. (2007), Exploring Chemical Analysis, W.H. Freeman and Co.

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
2. **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit**(of **19 marks**) will consist of two types of questions:**9 multiple-choice questions**of one mark each and**5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the compulsory question and one question each from UNIT-1, UNIT-2 and UNIT-3.
2. The use of scientific calculators is not allowed.

Semester-1
Syllabus for Practical Paper
ICH-DSC-1-MAJ/MIN-101
Basic Industrial Principles, Data Analysis and Laboratory Operations

End semester = 22	**Credit - 01**
Internal Marks = 03	**Contact Hours – 3/week**
Examination Time allowed	**3 Hours**

Basic Industrial Laboratory Operations:

1. Use and calibration of volumetric equipment and glassware.
2. Preparation of standard solutions of NaOH, $KMnO_4$, H_2SO_4, HCl, Na_2CO_3, Oxalic acid. Introduction to primary and secondary standards.
3. Preparation of pure sample of potash alum.

Data Analysis and Laboratory Operations:

4. To plot accuracy graphfor each value of the independent variable using:
   (a) Excel data sheet (b) Origin software
5. To determine the errors in experimental data by plotting error bar graphs using:
   (a) Excel data sheet (b) Origin software
6. Calibration of equipment.
7. Preparation of buffer solutions.
8. Use of pH meter: determination of pH of given solutions.
9. Power point presentations.

Books Recommended

1. Seamus P.J. Higson: Analytical Chemistry.
2. Douglas A. Skoog and Donald M.West: Fundamentals of Analytical Chemistry.
3. Adion A. Gordus: Schaum's Outline of Analytical Chemistry, Tata McGraw-Hill.
4. Gary D. Christian: Analytical Chemistry.
5. Freifelder and Kealy: Analytical Chemistry.
6. Daniel C Harris: Exploring Chemical Analysis.
7. Daniel C Harris: Quantitative Chemical Analysis.

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (i.e. half hour for written and next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written+5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.

Semester	**II**
Code of Paper	**ICH-DSC-2-MAJ/MIN-201**
Title of Paper	**Applied Industrial Chemistry**
Total Credit	**04 (03 Theory + 01 Practical)**
Total Marks	**100(75Theory + 25 Practical)**
Marks distribution of the Theory paper	**End semester = 67 Internal Marks = 08**
Marks distribution of the Practical paper	**End semester = 22 Internal Marks = 03**
Contact Hours	**Theory: 3 Hrs. per week Practical: 3 Hrs. per week**
Applicable to the students **1. Those who have opted for Industrial chemistry as a Major subject in Table 2.** **2. Those who have opted for Industrial chemistry as a Minor subject in Table 1 or Table 2.**	
LEARNING OUTCOMES SPECIFIC TO DISCIPLINE By studying this applied course, the students will be able to: • Explain the details of fats, oils, soap and paper. • The student shall be able to give the details involved in paper manufacture. • Explain the working of sugar, cement, cosmetic, perfume and ceramic industries. • Emphasis is given to developing practical skills that will aid the student to set up accurate and reproducible, result-oriented experiments in higher classes.	
GENERIC OUTCOMES The first year of the course is designed to heighten the chemical temperament amongst the students. This is done by teaching them how to collect samples/data/evidence, carry out data analysis, evaluation and draw logical conclusions. They are also encouraged to work and improve upon their communication skills, through proper listening, plenty of hands-on laboratory exposure, group discussions and individual expression.	

For practical work, groups shall be constituted, with student strength not exceeding 20 students per group.

Semester 2: TRAINING/WORKSHOP/INDUSTRIAL VISITS

1. Visit to a local industry during the year and students to submit a report on the same. The report should contain information about: Raw materials used in industry, Steps involved in manufacturing, Final product and quality control measures involved.
2. Each student to deliver a seminar/power point presentation relevant to theory /practical syllabus.

Semester-2
Syllabus for Theory Paper
ICH-DSC-2-MAJ/MIN-201
Applied Industrial Chemistry

End semester = 67	**Credits - 03**
Internal Marks = 08	**Contact Hours – 3/week**
Examination Time allowed	**3 Hours**

UNIT-1(15 Hrs.)

Oils & Fats: Introduction, Chemistry of oils and fats, including their structure, composition, and classification, Saturated and unsaturated oils/fats, Saturated and unsaturated fatty acids with example, Physical and chemical characteristics of oils/fats, Extraction of oils from vegetable oils, refining of oils and fats, hydrogenation of oils, Determination of quality parameters of oils/fats: acid value, saponification value and iodine value, Industrial applications.

Soaps and Detergents: Introduction, Theoretical aspects, Chemistry of soaps and detergents, Chemical composition, structures, and properties of both soaps and detergents, Types of soaps, Various raw materials used in the manufacture of soaps, such as oils, fats, alkalis, and other additives, Manufacture of soap: Cold, semi-boiled and hot process, Cleaning action of soap, Chemical reactions involved in the saponification process, mechanism of action of detergents., Classification of detergents, finishing of detergents, Comparison between Soaps and Detergents.

Pulp & paper: Introduction, Raw material preparation, pulping techniques (mechanical and chemical), types of pulp, Manufacture of paper: paper formation and finishing, Chemicals used in paper manufacturing.

UNIT-2(15 Hrs.)

Sugar: Introduction; Sugar industry overview, Chemicals used in sugar processing, Manufacturing of cane sugar, Steps involved in manufacturing, Sugar extraction process: defecation, sulphitation, carbonation, evaporation, concentration, crystallization, clarification, centrifugation, drying, refining; By-Product of sugar industry, Uses of molasses and bagasse.

Cement: Introduction to Cement: Definition, classification, and applications, Cement-Types of cement, Constituents of cement, Types of raw materials used in cement production, including limestone, clay, and gypsum, Manufacture of Portland cement, Flow chart of Portland cement manufacturing process, Chemical composition of Portland cement, Mixing and Crushing, Dry Process, Wet Process, Burning Process, Rotary Kiln zones, Grinding and Packaging, Setting and Hardening of Portland cement, Heat of hydration of cement.

Ceramics: Introduction, Definition, Properties of ceramics: Chemical, mechanical, thermal, Classification of ceramic materials: Oxides, non-oxides and composites, Traditional classification of ceramics: Pottery, Whiteware, Stoneware, Chinaware, Porcelain, Refractories, Classification of refractories: acid, basic and neutral, Characteristics of raw materials, Properties of refractories, Important steps involved in the manufacture of refractories.

UNIT-3(15 Hrs.)

Cosmetics and Perfumes: Chemistry of cosmetics and perfumes, Introduction to cosmetics and perfumes, Types of cosmetics, Ideal properties of cosmetics, Classification of cosmetics: according to their use, according to their function, according to their physical nature, according to function state.

General study including preparation and uses: Hair dye, hair spray, Shampoo.

Raw material, manufacturing and uses: Suntan lotions, face powder, lipsticks, talcum powder, nail enamel.

Preparation and uses: creams (cold, vanishing and shaving creams), antiperspirants and artificial fragrances.

Essential oils: Their importance in cosmetic industries with reference to; Eugenol, Geraniol, Sandalwood oil, Eucalyptus oils and Rose oil, their production and uses.

Books Recommended

1. Shreev, R.N. & Brink, J.A.: Chemical Process Industries, 5th Edition, McGraw Hill, 1987.
2. Austine, G.T.: Shreeves Chemicals Process Industries, 5th Edition, McGraw Hill, 1984.
3. Dryden, C.E., Rao M.G. & Silting, M.: Outlines of Chemical Technology, 3rd Edition, affiliated East West Press Pvt. Ltd., N. Delhi, 2008.
4. Pandey, G.N.: Chemical Technology, Volume-II, Lion Press, Kanpur.
5. Hench, L.L.: Science of Ceramic chemical processing.
6. Stewarts, G.H.: Science of Ceramics.

Instructions for the examiner

4. The examiner will set the question paper comprising **UNIT-1, UNIT-2, UNIT-3** and a **compulsory Unit**.
5. **UNIT-1, UNIT-2 and UNIT-3** will have two questions each of **16 marks**. Each question shall have 3-4 questions as sub-parts as sub-parts equally distributed within the syllabus of the respective unit.
6. The **compulsory unit**(of **19 marks**) will consist of two types of questions:**9 multiple-choice questions**of one mark each and**5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **four questions in all:** the compulsory question and one question each from UNIT-1, UNIT-2 and UNIT-3.
2. The use of scientific calculators is not allowed.

Semester-2
Syllabus for Practical Paper
ICH-DSC-2-MAJ/MIN-201
Applied Industrial Chemistry

End semester = 22	**Credit - 01**
Internal Marks = 03	**Contact Hours – 3/week**
Examination Time allowed	**3 Hours**

Chemistry Lab: Applied Industrial Laboratory Operations

1. Determination of Acid value of the given oil/fat material.
2. Determination of Iodine value of the given oil/fat material.
3. To determine Saponification value of the given oil/fat materials.
4. To make a recycled paper on laboratory scale.
5. Carbohydrates: Reducing and Non reducing sugars by:
   (a) Fehling's method (b) Pavy's method
6. Cement: Loss of ignition, silica, insolubles, ash content.
7. To determine the porosity of the given ceramic material.
8. Preparation of nail polish and nail polish remover.
9. Power point presentations.

Books Recommended

1. Vermani, O. P.; Narula, A. K. (2004), Industrial Chemistry, Galgotia Publications Pvt. Ltd., New Delhi.
2. Bhatia, S. C. (2004), Chemical Process Industries, Vol. I & II, CBS Publishers, New Delhi.
3. Barel, A.O.; Paye, M.; Maibach, H.I. (2014), Handbook of Cosmetic Science and Technology, CRC Press.
4. Gupta, P.K.; Gupta, S.K. (2011), Pharmaceutics and Cosmetics, Pragati Prakashan.
5. Butler, H. (2000), Poucher's Perfumes, Cosmetic and Soap, Springer.

General Instructions to the Examiners:

Note: Practical examination will be of three hours duration divided into two parts (i.e. half hour for written and next two and half hours for performance).

1. Candidate will be examined based on any two experiments given by the examiner. Each experiment carries 08 marks (3 written+5 performance).
2. Viva-Voce: 04 marks.
3. Note Book: 02 marks.

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MATHEMATICS

AS PER NEP (2020)

Syllabus of **Mathematics** for UG Classes

Table-1(SCIENCE TEMPLET)

SINGLE MAJOR, SINGLE / DOUBLE MINOR.**BSc,BScHONOURSANDBSc HONOURS WITH RESEARCH**

Sem.	Level	MAJOR (4Cr)	MINOR (4 Cr)	MDC(3 Cr) Any paper in each of the first three semesters	VAC(2Cr) Required three papers, one for each in 1st, 2nd, & 4th semesters	SEC(3Cr) Required three papers, one for each of the first three semesters	AEC (2Cr)	PBI / HCP (2Cr)
I	100	• MAT-DSC1 (MAJ/MIN)-101: Algebra and Trigonometry	• MAT-DSC1 (MAJ/MIN)-101: Algebra and Trigonometry	• MAT-MDC-101:BasicMathematics • MAT-MDC-102:Calculus • MAT-MDC-103:Algebra	• MAT-VAC-101:Vedic Mathematics • MAT-VAC-102:Vedic Algebra • MAT-VAC-103:Theory of Numbers • MAT-VAC-104:Theory of Numbers with html	• MAT-SEC-101:Maths Based Practicals • MAT-SEC-102:Maths Based Programming AndPracticals • MAT-SEC-103:Latex And Html • MAT-SEC-104:Quantitative Aptitude • MAT-SEC-105:Discrete Mathematics	Eng -1	Pbi-1/ HCP-1
II	100	• MAT-DSC2 (MAJ/MIN)-201: Calculus-I	• MAT-DSC2 (MAJ/MIN)-201: Calculus-I				Eng -2	Pbi-2/ HCP-2
III	200	• MAT-DSC3 (MAJ/MIN)-301: Theory of Equations • MAT-DSC3 (MAJ)-302: Differential Equations-I	• MAT-DSC3 (MAJ/MIN)-301: Theory of Equations • MAT-DSC3 (MAJ)-302: Differential Equations-I				Eng-3/ MIL-1	
IV	200	• MAT-DSC4 (MAJ/MIN)-401: Calculus-II • MAT-DSC4 (MAJ)-402: Discrete Mathematics and Graph Theory • MAT-DSC4 (MAJ)-403: Coordinate Geometry-I • MAT-DSC4 (MAJ)-404: Numerical Analysis	• MAT-DSC4 (MAJ/MIN)-401: Calculus-II				Eng-4/ MIL-2	
V	300	• MAT-DSC5 (MAJ/MIN)-501: Linear Algebra • MAT-DSC5 (MAJ)-502: Sequence and Series of real numbers • MAT-DSC5 (MAJ)-503: Differential Equations-II • MAT-DSC5 (MAJ)-504: Solid Geometry • MAT-DSC5 (MAJ)-505: Statics			Internship / community work			
VI	300	• MAT-DSC6 (MAJ/MIN)-602: Analysis • MAT-DSC6 (MAJ)-603: Linear Programming • MAT-DSC6 (MAJ)-604: Probability Theory • MAT-DSC6 (MAJ)-605: Dynamics • MAT-DSC6 (MAJ)-606: Modern Algebra	MAT-DSC6 (MAJ/MIN)-601: Analytical Geometry					

Table-2(SCIENCE TEMPLETE)

MULTI-DISCIPLINARY / INTER-DISCIPLINARY. BSc,BScHONOURSANDBSc HONOURS WITH RESEARCH

Sem.	Level	MAJOR (4 Cr) A	MAJOR (4 Cr) B	MINOR (4 Cr) Cz	MDC(3 Cr) Any paper in each of the first three semesters	VAC(2Cr) Required three papers, one for each in 1st, 2nd, & 4th semesters	SEC(3Cr) Required three papers, one for each of the first three semesters	AEC (2Cr)	PBI / HCP (2Cr)
I	100	• MAT-DSC1 (MAJ/MIN)-101: Algebra and Trigonometry	• MAT-DSC1 (MAJ/MIN)-101: Algebra and Trigonometry	• MAT-DSC1 (MAJ/MIN)-101: Algebra and Trigonometry	• MAT-MDC-101:BasicMathematics • MAT-MDC-102:Calculus • MAT-MDC-103:Algebra	• MAT-VAC-101:Vedic Mathematics • MAT-VAC-102:Vedic Algebra • MAT-VAC-103:Theory of Numbers • MAT-VAC-104:Theory of Numbers with html	• MAT-SEC-101:Maths Based Practicals • MAT-SEC-102:Maths Based Programming And Practicals • MAT-SEC-103:Latex And Html • MAT-SEC-104:Quantitative Aptitude • MAT-SEC-105:Discrete Mathematics	Eng -1	Pbi-1/ HCP-1
II	100	• MAT-DSC2 (MAJ/MIN)-201: Calculus-I	• MAT-DSC2 (MAJ/MIN)-201: Calculus-I	• MAT-DSC2 (MAJ/MIN)-201: Calculus-I				Eng -2	Pbi-2/ HCP-2
III	200	• MAT-DSC3 (MAJ/MIN)-301: Theory of Equations	• MAT-DSC3 (MAJ/MIN)-301: Theory of Equations	• MAT-DSC3 (MAJ/MIN)-301: Theory of Equations • MAT-DSC3 (MAJ)-302: Differential Equations-I				Eng-3/ MIL-1	
IV	200	• MAT-DSC4 (MAJ/MIN)-401: Calculus-II • MAT-DSC4 (MAJ)-402: Differential Equations-I	• MAT-DSC4 (MAJ/MIN)-401: Calculus-II MAT-DSC4 (MAJ)-402: Differential Equations-I	• MAT-DSC4 (MAJ/MIN)-401: Calculus-II				Eng-4/ MIL-2	
V	300	• MAT-DSC5 (MAJ/MIN)-501: Linear Algebra • MAT-DSC5 (MAJ)-502: Sequence and Series of real numbers • MAT-DSC5 (MAJ)-503: Differential Equations-II	• MAT-DSC5 (MAJ/MIN)-501: Linear Algebra	• MAT-DSC5 (MAJ/MIN)-501: Linear Algebra		Internship / community work			
VI	300	• MAT-DSC6 (MAJ/MIN)-601: Analytical Geometry • MAT-DSC6 (MAJ)-602: Analysis • MAT-DSC6 (MAJ)-603: Modern Algebra	• MAT-DSC6 (MAJ/MIN)-601: Analytical Geometry	• MAT-DSC6 (MAJ/MIN)-601: Analytical Geometry • MAT-DSC6 (MAJ)-602: Analysis					

Notes.

1. Members of the Board of Studies unanimously decided not to offer mathematics in Table-3 of Arts stream.

2.Practicals (wherever included in the syllabi) may be conducted into small groups(say 15 – 20 students)to facilitate the smooth conduct of practical sessions.

3. Scientific Calculator is allowed in the Numerical Analysis Paper.

IMPORTANT NOTE: All papers in VAC/SEC/MDC would be floated in each of the first three semesters. However, the students are not allowed to repeat any paper.

Syllabus as per Table1 and 2 (Sciences)

Semester - I

MAT-DSC1(MAJ/MIN)-101: ALGEBRA& TRIGONOMETRY

Credits = 4
Teaching hours = 6 hrs / week including tutorials /practicals etc. in a semester of 15 weeks.
Exam Time: 3 hours
Max. Marks: 100 (External: 80 + Internal Assessment: 20)

Note: 1. The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 4 parts covering the entire syllabus. Two questions will be set from each Unit.
2.A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.
3. The 20 marks of internal assessment has following division

Mid Semester Exam = 10 marks,
Assignment / Project work = 5 marks,
Classroom participation = 3 marks**,**
Attendance = 2 marks for more than 85%and 1mark for more than 75% attendance.

Objective:The main goal of this course is to deliver basics of Binomial Theorem, Properties of Binomialcoefficients.ThecourseintroducestheconceptofDeMoivre'stheoremanditsapplications,eigen values and eigen vectors of a matrix. The objective is to understand nature of solutionofalinearsystem ofequations.

Learning Outcomes: The students will be able to determine the sum of infinite Binomial Series, the rank ofamatrix,eigenvalues,eigenvectorsandsolvelinearsystemofequationsusingmatrices.

UNIT-I

Principle of Mathematical Induction (both strong and weak forms) and its applications (e.g. see [1]). Binomial Theorem for positive index, verification of its correctness by using induction principal, its combinatorial interpretation, Properties of binomial coefficients. Binomial Theorem for anyindex,summationofinfinite binomialseries.
Solutionoftrigonometricequations,sine, cosine and projectionformulaeforarbitrary triangles (e.g. see [3]).

UNIT-II

De Moivre's theorem, applications of De Moivre's theorem including primitive n^{th} roots of unity.Expansionsofsin$n\theta$,cos$n\theta$, $sin^n\theta$, $cos^n\theta$ for $n\in \mathbb{N}$.Exponential,logarithmic,circular (directandinverse) andhyperbolicfunctionsofacomplex variable.

UNIT –III

Determinantofan $n\times n$matrix and itsproperties. Definition and properties of hermitianandskew-hermitianmatrices. Rowand column vectors, linearly dependent and independent vectors, row rank, column rank and theirequivalence, rank of a matrix. Rank of product of matrices and rank of sum of matrices.

UNIT-IV

Theoremsonconsistencyofasystemoflinearequations(both homogeneousandnon-homogeneous). Eigen-values, eigen-vectors and characteristic equation of a matrix, Cayley-Hamiltontheoremanditsuse infinding inverse ofamatrix.Diagonalization.

References:

1. **I. S. Sominskii**: The Method of Mathematical Induction, Blaisdell Publishing Company New York. London-Pergamon Press Ltd (1961).
2. **David S. Gunderson**: Handbook of Mathematical Induction; Theory and Applications, CRC Press (2010).
3. **C. V. Durell**, **A. Robson**: Advanced Trigonometry, Dover Publications (2003).
4. **K.B.Datta**:MatrixandLinearAlgebra,PrenticeHallofIndiaPvt.Ltd.,NewDelhi,2004.
5. **S.R.KnightandH.S.Hall**:HigherAlgebra,H.M.Publications,SeventhEdition,1994.
6. **R.S.VermaandK.S.Shukla**:TextBookonTrigonometry,PothishalaPvt.Ltd.,Allahabad,10th Edition,1999.
7. **ShantiNarayanandP.K.Mittal**:ATextBookofMatrices,S.Chand&Co.,NewDelhi,RevisedEdition,2007.

Semester - II

MAT-DSC2(MAJ/MIN)-201: CALCULUS-I

Credits = 4
Teaching hours = 6 hrs / week including tutorials /practicals etc. in a semester of 15 weeks.
Exam Time: 3 hours
Max. Marks: 100 (External: 80 + Internal Assessment: 20)

Note: 1. The syllabus has been split into Four Units. The examiner will set in total nine questions including a compulsory question. The compulsory question (QuestionNo.1) will contain 4 parts covering the entire syllabus. Two questions will be set from each Unit.
2. A student will be required to attempt five questions in all, selecting one question from each Unit and the compulsory question. Each question will carry equal marks.
3. The 20 marks of internal assessment has following division
Mid Semester Exam = 10 marks,
Assignment / Project work = 5 marks,
Classroom participation = 3 marks**,**
Attendance = 2 marks for more than 85%and 1mark for more than 75% attendance.

Objective:The main goal of this course is to introduce the basics of limits, continuity and differentiability offunctions.Besidesthis,differentiationofhyperbolicandinversehyperbolicwiththeirapplications will also be studied. The course will cover Mean Value Theorems and theirapplications.

LearningOutcomes:Aftercompletingthiscoursethestudentswillbeabletocomputelimitsandcontinuityof variousfunctions. Thecoursewillempowerthestudentsto computederivativesofHyperbolicfunctions.Thestudents willbeabletocomputetangents andnormals.

UNIT-I

Real Numbers. Orderpropertiesofrealnumbers,bounds,*l.u.b.*and*g.l.b.*,ordercompletenesspropertyofrealnumbers,Archimedean property ofreal numbers (e.g. see chapter 1 of [1]).

Limits:Functions (exponential, logarithmic, modulus, trigonometric, polynomials etc.), ε- δ definitionofthelimitofafunction,basic propertiesoflimits, methods computations, infinitelimits (e.g. see chapter 5 of [1]).

UNIT-II

Continuity. ε- δ**definition of a continuous function, various methods to check continuity / discontinuity of a function, types of discontinuities, continuity of composite** functions, signof afunctioninaneighborhoodof apointof continuity,intermediatevaluetheorem,maximumandminimumvaluetheorem.

Differentiability. Definition of a differentiable real valued function of a real variable, computing derivatives of elementary functions by using definition. Geometrical meaning of derivative of a function at a point.

UNIT-III

Derivatives. Revision of various rules to compute derivatives (e.g. product rule, quotient rule, chain rule etc.), Introduction to hyperbolic,inversehyperbolicfunctionsofarealvariable, theirderivatives. Successive differentiations,Leibnitz'stheorem, indeterminateforms.

UNIT-IV

ApplicationsofDerivatives: TangentsandnormalsDifferentialsandApprox./Errors.

Meanvaluetheorems:Rolle'sTheorem,Lagrange'smeanvaluetheorem,Cauchy'smeanvaluetheorem,their geometricinterpretationandapplications,Taylor'stheorem,Maclaurin'stheoremwithvariousformofremainders andtheirapplications.

References

1. **Brian S. Thomson, Judith B. Bruckner, Andrew M. Bruckner:** Elementary real analysis, Prentice-Hall (2001)
2. **MurrayR.Spiegel**:Schaum'soutlineseriesTheoryandProblemsofAdvancedCalculus, SchaumPublishingCo.NewYork,SecondEdition,2002.
3. **P.K.JainandS.K.Kaushik**:AnIntroductiontoRealAnalysis,S.Chand&Co.,NewDelhi,First Edition,2000.
4. **G. Prasad**:Differential Calculus, Pothishala Pvt Ltd., Allahabad, Revised Edition,2016.
5. **G.B.ThomasandR.L.Finney**:CalculusandAnalyticGeometry,PearsonPublication,NinthEdition,2010.
6. **ShantiNarayanandP.K.Mittal**: DifferentialCalculus,S.Chand&Co.,NewDelhi,FifteenthEdition,1942.
7. **R.G.Bartle,D.R.Sherbert**:IntroductiontoRealAnalysis,WileyIndiaPvt.Ltd

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MICROBIOLOGY

NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR (Table 1*) and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY (Table 2*)

B.Sc., B.Sc. HONOURS AND B.Sc. HONOURS WITH RESEARCH IN MICROBIOLOGY

Table 1*						Table 2*					
Sem	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**	**Sem**	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**
I	L-100	(MIC-DSC-1 MIN - 101)	Basic Microbiology	3+1=4	24	I	L-106	(MIC-DSC-1 MIN - 101)	Basic Microbiology	3+1=4	24
		DSC- MAJ	To be opted by the student	3+1=4				DSC- MAJ	To be opted by the student	3+1=4	
		DSC- MAJ	To be opted by the student	3+1=4				DSC- MAJ	To be opted by the student	3+1=4	
		(MIC - MDC-1-101)	Introduction to Microbiology	3				(MIC - MDC-1-101)	Introduction to Microbiology	3	
		(VAC-1-101)	To be opted by the student	2				(VAC-1-101)	To be opted by the student	2	
		(MIC-SEC-1-101)	Tools and Techniques in Microbiology-I	3				(MIC-SEC-1-101)	Tools and Techniques in Microbiology-I	3	
		(AEC-1-101)	Eng-1	2				(AEC-1-101)	Eng-1	2	
		(PUN-COMPL- 1 – 101/HCP-COMPL-1-101)	Pbi-1/HCP-1	2				(PUN-COMPL-1 – 101/HCP-COMPL-1-101)	Pbi-1/HCP-1	2	
II	L-100	(MIC-DSC-2 MIN - 201)	Microbial Physiology	3+1=4	24	II	L-100	(MIC-DSC-2 MIN - 201)	Microbial Physiology	3+1=4	24
		DSC- MAJ	To be opted by the student	3+1=4				DSC- MAJ	To be opted by the student	3+1=4	
		DSC- MAJ	To be opted by the student	3+1=4				DSC- MAJ	To be opted by the student	3+1=4	
		(MIC-MDC-2-201)	Tools and Techniques in Microbiology-II	3				(MIC-MDC-2-201)	Tools and Techniques in Microbiology-II	3	
		(VAC-2-201)	To be opted by the student	2				(VAC-2-201)	To be opted by the student	2	
		(MIC-SEC-2-201)	Biosafety in Microbiology	3				(MIC-SEC-2-201)	Biosafety in Microbiology	3	
		(AEC-2-201)	Eng-2	2				(AEC-2-201)	Eng-2	2	
		(PUN-COMPL- 2 - 201/HCP-COMPL-2-201)	Pbi-2/HCP-2	2				(PUN-COMPL-2 - 201/HCP-COMPL-2-201)	Pbi-2/HCP-2	2	

The candidate has to select Major and Minor out of 3 subjects studied in the I and II Semester at the end of the Semester II

The student exiting the programme after one year, securing 48 credits, will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work.

***Note: The candidate will opt for either Template 1 or 2 from Semester III onwards.**

***The internship will consist of a 4-credit course (100 marks) comprising 30 hours (2 hours per week) of practical or field training. The internship may be completed within the parent institute or through institute/industrial training for 2–4 weeks, with a minimum requirement of 30 hours. Evaluation will be based on the project report and a detailed presentation of the internship. The internal assessment will constitute 10% of the total marks, as per university and institutional guidelines.**

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

SEMESTER-I

MIC-DSC-1 MIN – 101: BASIC MICROBIOLOGY (THEORY)

Total marks: 75
Theory: 68
Internal assessment: 07
Time: 3 hours
Credits: 3

Instructions for paper setters and candidates:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit.
- Compulsory question will be of 12 marks. Rest all questions will be of 14 marks each

Objective And Learning outcome:

Aim of the course is to introduce the students with microbial world and to make the students to learn the morphology, microscopy, staining. It will also enrich the students' knowledge on the development of microbiology as the discipline.

UNIT–I

NO. OF HOURS: 15

HISTORY OF MICROBIOLOGY

Development of microbiology as a discipline, Spontaneous generation vs. biogenesis. Contributions of Antoni van Leeuwenhoek, Louis Pasteur, Robert Koch, Joseph Lister, Alexander Fleming. Germ theory of disease. Overview of classification of microbes

MICROSCOPY

Principle and applications of bright field microscopy, dark field microscopy, phase contrast microscopy and electron microscopy.

STAINING

Nature of stains, principles, mechanism, methods and types of staining: Simple, Negative staining, Gram staining, Capsule staining and Acid fast staining.

UNIT–II

NO.OF HOURS: 15

MICROBIAL CELL STRUCTURE AND ORGANIZATION

Prokaryotic vs Eukaryotic cell organization,
Bacterial cell structure Cell size shape and arrangement, Organization of cell wall, cell membrane, flagella and capsules, fimbriae and pili, mesosomes, nucleoid chromosome and plasmids in bacteria.

BACTERIOLOGICAL TECHNIQUES

Pure culture isolation: Culture media, Streaking, serial dilution and planting methods; cultivation, maintenance and preservation/stocking of pure cultures; cultivation of anaerobic bacteria.

UNIT-III

NO. OF HOURS:15

MICROBIAL GROWTH

Phases of growth, generation time, growth rate, bacterial growth curve, mathematical nature and expression, diauxic growth, chemostat and turbidostat.
Factors affecting growth in microorganisms.
Endospore structure formation and significance.
Methods for measurement of growth in microorganisms.

UNIT-IV

NO.OF HOURS:15

CONTROL OF MICROORGANISMS

Physical (Heat, radiation, filtration) and chemical methods of sterilization (disinfection, antiseptics), antimicrobial agents, sulfa drugs, antibiotics: broad-spectrum antibiotics, beta-lactam antibiotics.

NORMAL MICROBIAL FLORA

Germfree and gnotobiotic life, distribution and occurrence of the normal flora: skin, eye, respiratory tract, mouth, intestinal tract, genitourinary tract.

SUGGESTED READINGS

1. Black JG. (2008). Microbiology: Principles and Explorations. 7th edition. Prentice Hall.
2. Madigan MT, and Martinko JM. (2014). Brock Biology of Micro-organisms. 14thedition. Parker J. Prentice Hall International, Inc.
4. Pelczar Jr MJ, Chan ECS, and Krieg NR. (2004). Microbiology. 5th edition Tata McGraw Hill.
5. Srivastava S and Srivastava PS. (2003). Understanding Bacteria. Kluwer Academic Publishers, Dordrecht.
6. Stanier RY,Ingraham JL, Wheelis ML and Painter PR. (2005). General Microbiology. 5th edition McMillan.
7. Tortora GJ, Funke BR, and Case CL. (2008). Microbiology: An Introduction. 9th edition Pearson Education.
8. Willey JM,Sherwood LM, and Woolverton CJ. (2013). Prescott's Microbiology. 9th edition. McGraw Hill Higher Education.
9. Cappucino J and Sherman N.(2010).Microbiology: A Laboratory Manual 9thedition .Pearson Publication

MIC-DSC-1 MIN – 101: BASIC MICROBIOLOGY (PRACTICALS)

Total marks: 25
Practical: 22
Internal assessment: 03
Time: 3 hours
Credits :1

List of the Practical:

1. Preparation of different media: Synthetic Media, Complex media.
2. Simple staining and Negative staining..
3. Gram staining
4. Capsule staining.
5. Isolation of pure cultures of bacteria by streaking method.
6. Estimation of CFU count by spread plate method/pour plate method.
7. Demonstration of motility by hanging drop method.

MULTIDISCIPLINARY

MIC-MDC-1-101: INTRODUCTION TO MICROBIOLOGY

Total marks: 50
Theory: 45
Internal assessment: 05
Time: 3 hours
Credits: 02

Note: The question paper will consist of Two Units. There will be nine questions, out of which five questions have to be attempted. Question I will span the complete syllabus and will be compulsory. Rest of the eight questions will be from different Units of the syllabus. There will be four questions from each Unit and two is to be attempted from each unit

Objective And Learning outcome: Aim of the course is to introduce the students with the microbes .Microbes play a significant role in medical science and industries .To study the microbes from basic, with an understanding of cell structure, and application of microbes.

UNIT–I

HISTORY AND SCOPE OF MICROBIOLOGY

General introduction and overview of microbial world. Major division of life. Members of microbial world and their characteristics.

Morphology of Bacteria, viruses, viroids and Prions, fungi, Algae, protozoa.

Development of techniques for studying microbial pathogens. The scope and relevance of microbiology.

MICROBIOTA AND HOST RELATIONSHIP

Distribution and role of Normal microflora of human Body.

Role of microorganisms in diseases. Microbial interactions like symbiosis and antibiosis.

UNIT-II

MICROSCOPY

Principles of microscopy; resolution, magnification; compound microscope. Principle and applications of Bright field microscopy, Dark field microscopy.

Different phases of growth curve, specific growth, doubling time. The influence of environmental factors on growth of microorganisms.

MEDIA PREPARATION

Culture media: Different types of media preparation methods. Sterilization and pasteurization; The Physical and chemical methods of sterilization, pasteurization

SUGGESTED READING

1. Willey, J. M., Sherwood, L. M., & Woolverton, C. J. (2014). *Prescott's microbiology*. McGraw-Hill.
2. Black JG. (2008). Microbiology: Principles and Explorations. 7th edition. Prentice Hall
3. Pelczar Jr MJ, Chan ECS, and Krieg NR. (2004). Microbiology. 5th edition Tata McGraw Hill.
4. Srivastava S and Srivastava PS. (2003). Understanding Bacteria. Kluwer Academic Publishers, Dordrecht.
5. Stanier RY,Ingraham JL, Wheelis ML and Painter PR. (2005). General Microbiology. 5th edition McMillan.
6. Tortora, G.J., Funke, B.R. and Case, C.L. (2009) Microbiology: An introduction (Benjamin/ Cummings publishing company, Inc).
7. Stanier R Y, M. Doudoroff, E. A. Adelberg (1999). General microbiology (MacMillian press London).
8. Willey J M, Sherwood LM, and Woolverton CJ. (2013). Prescott's Microbiology. 9th edition. McGraw Hill Higher Education.
9. Postgate, J.R. (2000). Microbes and man (Cambridge University Press).
10. Alcamo, I. E. (2003). *Microbes and society: an introduction to microbiology*. Jones & Bartlett Learning.
11. Cappucino J and Sherman N.(2010).Microbiology: A Laboratory Manual9th edition. Pearson Publication

MIC-MDC-1-101: INTRODUCTION TO MICROBIOLOGY (PRACTICAL)

Total: 25
Practical: 22
INTERNAL ASSESSMENT:03
TIME: 3 hours
CREDITS:1

List of the Practical:

1. Basic Laboratory Rules and Instruments used in microbiology lab.
2. The components and use of compound microscope.
3. Preparation and sterilization of culture media.
4. Microscopic examination of microorganism by Temporary Wet Mount.
5. Measurements of microorganisms of the compound
6. Demonstration of Normal Microbial flora of the skin.
7. Isolation of microorganism using streak and pour plating method.
8. To perform simple staining.

SEMESTER-II

MIC-DSC-2 MIN-201: MICROBIAL PHYSIOLOGY (THEORY)

Total marks: 75
Theory: 68
Internal Assessment: 07
Time: 3 hours
Credits:3

Instructions for paper setters and candidates:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit.
- Compulsory question will be of 12 marks. Rest all questions will be of 14 marks each

Objectives and Learning outcome: The course is designed to understand physiological processes of microorganisms, study microbial nutrition, metabolism, and energy generation, explain microbial growth, regulation, and stress responses, introduce adaptations to extreme environment. After completing this course, students will be able to Explain nutritional and metabolic diversity of microorganisms. Understand mechanisms of nutrient uptake and transport, understand energy-yielding pathways in microbes
and analyze factors affecting microbial growth and survival, Correlate microbial physiology with industrial and environmental applications

UNIT –I **15 HOURS**

Microbial Nutrition & Transport Systems
Nutritional types of microorganisms, Macronutrients and micronutrients, Growth factors and vitamins
Nutrient uptake mechanisms: Passive diffusion, Facilitated diffusion, Active transport, Group translocation
Iron uptake and siderophores

UNIT –II **15 HOURS**

Microbial Metabolism & Bioenergetics
Concepts of metabolism: catabolism and anabolism
ATP generation and energy coupling, Oxidation–reduction reactions, Electron transport chain in bacteria
Chemiosmotic theory

UNIT –III **15 HOURS**

Carbohydrate, Lipid & Protein Metabolism
Glycolysis (EMP pathway), Entner–Doudoroff pathway, Pentose phosphate pathway
Fermentation types (alcoholic, lactic acid, mixed acid)
Lipid metabolism: β-oxidation
Overview of Amino acid metabolism and deamination

UNIT–IV **15 HOURS**

Microbial Regulation & Adaptation
Regulation of metabolism: induction and repression, Catabolite repression
Stress responses: heat shock, oxidative stress, osmotic stress
Physiological Adaptations.
Aerobic and anaerobic respiration
Photosynthetic microorganisms
Nitrogen fixation
Sulfur and phosphorus metabolism
Extremophiles: thermophiles, halophiles, acidophiles

SUGGESTED READINGS:

1. Berg JM, Tymoczko JL and Stryer L (2011) Biochemistry, W.H.Freeman and Company Caldwell, D.R. (1995). Microbial Physiology and Metabolism, W.C. Brown Publications,Iowa, USA.
2. Sashidhara Rao, B. and Deshpande, V. (2007). Experimental Biochemistry: A student Companion. I.K. International Pvt. Ltd.
3. Tymoczko JL, Berg JM and Stryer L (2012) Biochemistry: A short course, 2nd ed.,
4. W.H.Freeman 6. Voet,D. and Voet J.G (2004) Biochemistry 3rd edition, John Wiley and Sons
5. Fisher, M. (2001). Lehninger principles of biochemistry, ; by David L. Nelson and Michael M. Cox. The Chemical Educator, 6, 69-70

MIC-DSC-2 MIN-201: MICROBIAL PHYSIOLOGY (PRACTICALS)

MAX. MARKS: 25
PRACTICAL: 22
INTERNAL ASSESSMENT: 03
TIME: 3 HRS.
CREDITS:1

1. Preparation of minimal and enriched media.
2. Effect of temperature on microbial growth.
3. Effect of Carbon/nitrogen source on microbial growth.
4. Growth curve determination using spectrophotometer.
5. Oxidase and catalase test.
6. Carbohydrate fermentation test with gas production
7. To demonstrate Aerobic vs anaerobic growth.

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PHYSICS

NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR(Table 1*)
and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY(Table 2*)
B.Sc., B.Sc. HONOURS AND B.Sc. HONOURS WITH RESEARCH

Table 1*						Table 2*					
Sem	Level	Course Code	Course Name	Credits (Th+Pr+Tut)	Total Credits	Sem	Level	Course Code	Course Name	Credits (Th+Pr+Tut)	Total Credits
I	L-100	(PHY-DSC-1 Maj/Min-101)	Mechanics	3+1=4 (x3)	24	I	L-100	(PHY-DSC-1 Maj/Min-101)	Mechanics	3+1=4 (x3)	24
		(PHY-MDC-1-101)	Green Energy	3				(PHY-MDC-1-101)	Green Energy	3	
		(PHY-VAC-1-101)	Science in Everyday Wonders	2				(PHY-VAC-1-101)	Science in Everyday Wonders	2	
		(PHY-SEC-1-101)	Physics instrumentation Skills-I	3				(PHY-SEC-1-101)	Physics instrumentation Skills-I	3	
		(AEC-1-101)	Eng-1	2				(AEC-1-101)	Eng-1	2	
		(PUN-COMPL1-101/HCP-COMPL1-101)	Pbi-1/HCP-1	2				(PUN-COMPL1-101/HCP-COMPL1-101)	Pbi-1/HCP-1	2	
II	L-100	(PHY-DSC-2 Maj/Min-201)	Electricity and Magnetism	3+1=4 (x3)	24	II	L-100	(PHY-DSC-2 Maj/Min-201)	Electricity and Magnetism	3+1=4 (x3)	24
		(PHY-MDC-2-201)	Principles of Motion and Healthy Living	3				(PHY-MDC-2-201)	Principles of Motion and Healthy Living	3	
		(PHY -VAC-2-201)	Fundamentals of Solar Energy and Applications	2				(PHY -VAC-2-201)	Fundamentals of Solar Energy and Applications	2	
		(PHY -SEC-2-201)	Physics instrumentation Skills-II	3				(PHY -SEC-2-201)	Physics instrumentation Skills-II	3	
		(AEC-2-201)	Eng-2	2				(AEC-2-201)	Eng-2	2	
		(PUN-COMPL2-201/HCP-COMPL2-201)	Pbi-2/HCP-2	2				(PUN-COMPL2-201/HCP-COMPL2-201)	Pbi-2/HCP-2	2	
The candidate has to select Major and Minor out of 3 subjects studied in the I and II Semester at the end of the II Semester											
The student exiting the programme after oneyear, securing 48 credits, will be awarded UG Certificate provided he/she secures 4credits from summer internship/community work.											

***Note: The candidate will opt for either Template 1 or 2 from Semester III onwards.**

Notes:

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain atleast 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

B.Sc. Semester 1
PHY-DSC-1 Maj/Min-101
(Mechanics)

Maximum marks: 100
(Theory 75 marks & Practical 25 marks)
Theory 75 marks [60 marks+15 marks (internal assessment 20%)]
Practical 25 marks [20 marks+5 marks (internal assessment 20%)]
Teaching hours: Theory- 3Hrs per week
Total Credits: 3 (Theory) + 1 (Practical)

Learning Outcomes

- To develop a comprehensive understanding of Cartesian and spherical polar coordinate systems.
- To establish the relationship between the conservation laws and the symmetries of space and time.
- To explore elastic collisions in laboratory and center-of-mass reference frames and to derive relationships between velocities, angles, and kinetic energies in these frames.
- To understand Galilean transformations, invariance, and the transformation equations for inertial frames.
- To investigate the postulates of the special theory of relativity and explore the kinematical consequences of Lorentz transformations, including length contraction, time dilation, and the twin paradox.
- To derive and understand the relativistic transformation of velocities, the concept of simultaneity in relativity, and the relativistic Doppler effect.

Note:

1. The theory paper will be of 75 marks including 15 marks for internal assessment and three Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

UNIT-I

Cartesian, plane polar and spherical polar co-ordinate systems, area, volume, displacement, velocity, and acceleration in these systems, Relationship of conservation laws of linear momentum, angular momentum and energy, symmetries of space and time.

Various forces in nature, relative strengths and spatial dependence, Motion under force obeying inverse square law, equivalent one-body problem. Motion under central forces, Equation of motion under central force, equation of orbit and turning points, Kepler's Laws, Elastic collision in lab and C.M. systems, relationships of velocities, angles, and kinetic energies in these two systems.

UNIT-II

Galilean transformations and Invariance, Non-Inertial frames, Fictitious forces in a rotating frame of reference, Centrifugal and Coriolis forces due to rotation of earth, Concept of stationery universal frame of reference and ether, Michelson-Morley experiment, and its results.

Postulates of special theory of relativity, Lorentz transformations, Kinematical consequences of Lorentz transformations – length contraction and time dilation, Twin paradox, Transformation of velocities, Relativistic Doppler effect, Variation of mass with velocity, mass-energy equivalence.

Books Suggested:

Essential Readings:

1. Mechanics, Berkeley, Vol. I, C. Kittle.
2. Mechanics, H.S. Hans & S.P. Puri.

Further Readings:

1. An Introduction to Machines, Daniel Kleppner & Robert J. Kolenkow (TMH).
2. Introduction of Classical Mechanics, R.G. Takwale & P.S. Puranik (TMH, 2000).
3. Mechanics & Relativity (3rd Edition) Vidwan Singh Soni (PHI Learning, New Delhi, 2013)
4. Basic Concepts of Relativity, R.H. Good (East-West Press, New Delhi, 1974).

PHYSICS PRACTICALS
(PHY-DSC-1 Maj/Min-101)

Max. Marks: 25 (20+5 internal assessment) **Total Credit: 1**

Teaching hours: 2 Hrs per week

General Guidelines for Physics Practical Examinations:

1. The distribution of marks is as follows:

(i)	One full experiment requiring the student to take some data, analyze it and draw conclusions. (Candidates are expected to state their results with limits of error).	09
(ii)	One exercise based on experiment	04
(iii)	Viva-Voce	04
(iv)	Record (Practical file)	03
(v)	Internal Assessment	05

Note for Examiners:

2. The marks scored under each head must be clearly written on the answer sheet.
3. There will be one session of 3 hours duration. The paper will have two sections. In Section-A one experiment is to be allotted by the external examiner.
4. Section–B will consist of exercises which will be set by the external examiner on the spot. The length of the exercises should be such that any of these could be completed in one hour.
5. The examiner should take care that the experiment allotted to an examinee from section–A and exercise allotted from section–B are not directly related to each other.
6. Number of candidates in a group for practical examination should not exceed 12.
7. In a single group, no experiment to be allotted to more than three examinees in the group.

Note: Each student will perform at least five experiments in the whole semester.

Analysis of Experimental Data:

Objectives:

(i) Knowledge of propagation of errors.

(ii) Knowledge of significant figures, Determination of standard deviation and probable error and their use in interpretation of the experimental result.

(iii) Familiarity with the method of least square fitting of experimental data to a curve.

LIST OF EXPERIMENTS:

1. Basics of measurements with Vernier calliper, screw gauge, spherometer, travelling microscope.
2. To find out the moment of inertia of a flywheel.
3. To establish relationship between torque and angular acceleration using fly wheel.
4. To study one dimensional elastic collision using two hanging spheres.
5. To determine energy transfer, coefficient of restitution and verify laws of conservation of linear momentum and kinetic energy in elastic collisions using one dimensional collision of hanging spheres.
6. To find Young's Modulus of the material of the given beam by bending using a travelling microscope
7. To Determine Poisson's ratio for rubber.
8. Determination of coefficient of viscosity of a given liquid by Stoke's method.

Texts and Reference Books:

1. "B.Sc. Practical Physics by C.L. Arora, S. Chand & Co. (2014).
2. "A Laboratory Manual of Physics for Undergraduate Classes" by D.P. Khandelwal.

B.Sc. Semester 2
PHY-DSC-2 Maj/Min-201
(Electricity and Magnetism)

Maximum marks: 100
(Theory 75 marks & Practical 25 marks)
Theory 75 marks [60 marks+15 marks (internal assessment 20%)]
Practical 25 marks [20 marks+05 marks (internal assessment 20%)]
Teaching hours: Theory- 3 Hrs per week
Total Credits: 3(Theory) + 1(Practical)

Learning Outcomes

- To gain a solid foundation in vector calculus, including the concepts of gradient, divergence, and curl in Cartesian coordinates.
- To study the fundamental principles of electrostatics, including Coulomb's law, electric fields from various charge distributions, and the concepts of electric potential and work.
- To explore the behavior of materials in electric fields, focusing on polarization, dielectric properties, and the relationships between electric susceptibility, permittivity, and the displacement vector.
- To study the behavior of electric currents, including current density and the microscopic form of Ohm's law.
- To understand the response of materials to magnetic fields, including the concepts of magnetization, magnetic susceptibility, permeability, and the relationships between free and bound currents.

Note:

1. The theory paper will be of 75 marks including 15 marks for internal assessment and three Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination Centre.
5. The Logarithmic table will be provided in the examination centre, if required.

UNIT-I

Basic ideas of Vector Calculus, Del Operator, Gradient, Divergence, curl in Cartesian coordinates, physical significance and applications, Conservative field, Laplacian in Rectangular coordinates. Stoke's theorem, Gauss's divergence theorem, Coulomb's Law for point charges and continuous distribution of charges, electric field due to dipole, line charge, charged ring, Gauss's Law and its differential form, Applications of Gauss's Law (Uniformly charge infinite wire, spherical shell and solid sphere).

Work and potential difference, Potential difference as line integral of field, Electric potential due to uniformly charged wire, dipole, linear quadrupole, Poisson and Laplace's equation,

Polarization of matter, atomic and molecular dipoles, induced dipole moment and atomic polarizability, Electric susceptibility and polarization vector, Relation K=1+ |, Gauss's law for dielectrics.

UNIT-II

Current density and equation of continuity, Microscopic form of Ohm's Law ($\mathbf{J} = \sigma\mathbf{E}$), Invariance of charge, **E** in different frames of reference.

Behaviour of various substances in magnetic field, Definition of M and H, Permeability and susceptibility and their inter relationship.

Lorentz's force, Definition of **B**, Biot Savart's Law and its application to long straight wire, circular current loop and solenoid, Ampere's Circuital law and its applications. Divergence and curl of **B**. Faraday's Law of EM induction, Self and Mutual inductance and reciprocity theorem.

Books Suggested:

Essential Readings:

1. Electricity & Magnetism, A.S. Mahajan & A.A. Rangwala (Tata McGraw Hill).
2. Fundamentals of Electricity and Magnetism by Arthur F. Kipp.
3. Electricity and Magnetism, Berkeley Physics Course, Vol. II by E.M. Purcell.
4. Introduction to Classical Electrodynamics by David Griffith, Prentice Hall.

Further Readings:

1. Electricity & Magnetism, 4th Edition, W.J. Duffin.
2. EM Waves and Radiating Systems, Edward C. Jordan and K. G. Balmain, Prentice Hall.

PHYSICS PRACTICALS
(PHY-DSC-2 Maj/Min-201)

Max. Marks: 25 (20+5 internal assessment) **Total Credit: 1**

Teaching hours: 2Hrs per week

General Guidelines for Physics Practical Examinations:

1. The distribution of marks is as follows:

(i)	One full experiment requiring the student to take some data, analyze it and draw conclusions. (Candidates are expected to state their results with limits of error).	09
(ii)	One exercise based on experiment	04
(iii)	Viva-Voce	04
(iv)	Record (Practical file)	03
(v)	Internal Assessment	05

Note for Examiners:

2. The marks scored under each head must be clearly written on the answer sheet.
3. There will be one session of 3 hours duration. The paper will have two sections. In Section-A one experiment is to be allotted by the external examiner.
4. Section–B will consist of exercises which will be set by the external examiner on the spot. The length of the exercises should be such that any of these could be completed in one hour.
5. The examiner should take care that the experiment allotted to an examinee from section–A and exercise allotted from section–B are not directly related to each other.
6. Number of candidates in a group for practical examination should not exceed 12.
7. In a single group, no experiment to be allotted to more than three examinees in the group.

Note: Each student will perform at least five experiments in the whole semester.

LIST OF EXPERIMENTS:

1. To study the working of household energy meter.
2. To study the efficiency of an electric kettle or heater element at constant input voltage.
3. To study the induced e.m.f. as a function of the velocity of the magnet.
4. To determine the capacitance of a capacitor using flashing and quenching of a neon lamp.
5. To study the magnetic field produced by a current carrying solenoid using a search coil and calculate permeability of air.
6. Study of phase relationships using impedance triangle for LCR circuit.
7. To study the resonance in series LCR circuits (for different R values) and calculate Q-value.
8. To study the resonance in parallel LCR circuits (for different R values) and calculate Q-value.
9. To determine the value of air capacitance by de-Sauty's method and hence to find out the permittivity of air.
10. To determine low resistance with Carey Fosters Bridge.

Texts and Reference Books:

1. "B.Sc. Practical Physics by C.L. Arora, S. Chand& Co. (2014).
2. "A Laboratory Manual of Physics for Undergraduate Classes" by D.P. Khandelwal.

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STATISTICS

NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR (Table 1*) and STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY(Table 2*)

B.Sc., B.Sc. HONOURS AND B.Sc. HONOURS WITH RESEARCH

Table 1*						Table 2*					
Sem	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**	**Sem**	**Level**	**Course Code**	**Course Name**	**Credits (Th+Pr+Tut)**	**Total Credits**
I	L-100	(STAT-DSC-1 Maj/Min-101)	Descriptive Statistics-I	3+1=4 (x3)	24	I	L-100	(STAT-DSC-1 Maj/Min-101)	Descriptive Statistics-I	3+1=4 (x3)	24
		(STAT-MDC-1-101)	Statistics-I	3				(STAT-MDC-1-101)	Statistics-I	3	
		(VAC-1-101)*	--	2				(VAC-1-101)*	--	2	
		(STAT-SEC-1-101)	Statistical Data Analysis using MS-Excel	2+1=3				(STAT-SEC-1-101)	Statistical Data Analysis using MS-Excel	2+1=3	
		(AEC-1-101)	Eng-1	2				(AEC-1-101)	Eng-1	2	
		(Pbi/HCP-1-101)	Pbi-1/HCP-1	2				(Pbi/HCP-1-101)	Pbi-1/HCP-1	2	
II	L-100	(STAT-DSC-2 Maj/Min-201)	Probabiltiy Theory	3+1=4 (x3)	24	II	L-100	(STAT-DSC-2 Maj/Min-201)	Probabiltiy Theory	3+1=4 (x3)	24
		(STAT-MDC-2-201)	Statistics-II	3				(STAT-MDC-2-201)	Statistics-II	3	
		(VAC-2-201)*	---	2				(VAC-2-201)*	---	2	
		(STAT-SEC-2-201)	Statistics for Decision Analytics with Microsoft Excel	2+1=3				(STAT-SEC-2-201)	Statistics for Decision Analytics with Microsoft Excel	2+1=3	
		(AEC-2-201)	Eng-2	2				(AEC-2-201)	Eng-2	2	
		(Pbi/HCP-2-201)	Pbi-2/HCP-2	2				(Pbi/HCP-2-201)	Pbi-2/HCP-2	2	
The candidate has to select Major and Minor out of 3 subjects studied in the I and II Semester at the end of the II Semester											
The student exiting the programme after oneyear, securing 48 credits, will be awarded UG Certificate provided he/she secures 4credits from summer internship/community work.											
***Value Added Courses are not being offered by the Deaprtment of Statistics, Panjab University, Chandigarh**											

***Note: The candidate will opt for either Template 1 or 2 from Semester III onwards.**

Semester-I

Course Code: (STAT-DSC-1 Maj/Min-101)
Course Name: Descriptive Statistics- I

Credit: 4
L-3, P-1

Notes:

1. A candidate shall offer this subject in B.Sc. (Hons) only if he/she studied Mathematics as a subject at 10+2 level.
2. Students are to be engaged to 5 hours of teaching per week (3 lectures for theory and 2 lectures for practical).
3. Practical examination will be held in the respective semester.

Paper Title with code	Lecture(L)/ Credit	Practical(P)/ Credit	Marks Distribution	
Descriptive Statistics-I (STAT- DSC- 1 Maj/ Min- 101)	Lecture :3 Credit: 3	Practical Lecture:2 Credit:1	Theory	Practical
			Max. Marks: 75 Theory: 60 Internal Assessment : 15 Time: 3 Hours	Max. Marks: 25 Viva voce: 5 Record of the practical: 5 Written practical Exam: 15 Time: 3 Hours

DESCRIPTIVE STATISTICS-I

Objective: The objective of the course is to Gain proficiency in presenting data through frequency distribution, diagrams, and graphical representations. To apply descriptive statistical techniques/methods to real-world datasets and draw meaningful insights.The focus will be both on theoretical as well as practical approach based on descriptive statistics.

Learning Outcomes: On completion of the course, the student will be able to:

1. Understand the origin, development, scope and importance of statistics.
2. Understand the nature of statistical data.
3. Identify the need of Classification and Tabulation.
4. Understand moments and their relationships, including their effects on change of origin and scale.
5. Construct frequency tables, interprets the data, and identifies the importance of diagrammatic and graphical presentation of data.
6. Explain, evaluate and interpret various measures of central tendency, dispersion, skewness and kurtosis.

Note: There will be nine questions in all. Question No.1 will be compulsory covering whole of the syllabus and comprising 5 to 7 short answer type questions. Rest of the eight questions will be set from the four units uniformly i. e. two from each unit. The candidate will be required to attempt five questions in all selecting one question from each unit and the compulsory one. All the questions will carry equal marks.

UNIT-I

Introduction of Statistics : Origin, development, definition, scope, uses and limitations.

Types of Data: Qualitative and quantitative data, nominal and ordinal data, time series data, discrete and continuous data, frequency and non-frequency data.

Collection and Scrutiny of Data: Collection of primary and secondary data-its major sources including some government publications, scrutiny of data for internal consistency and detection of errors of recording, classification and tabulation of data.

UNIT-II

Presentation of Data: Frequency distribution and cumulative frequency distribution, diagrammatic and graphical presentation of data, construction of bar, pie diagrams, histograms, frequency polygon, frequency curve,ogives, stem-and-leaf-chart, Box and whisker plot.

Measures of Central Tendency or Averages : Concepts of central tendency, characteristics for an ideal measure of central tendency, arithmetic mean, median, mode, geometric mean, harmonic mean; partition values-quartiles, deciles, percentiles and their graphical location along with their properties, applications, merits and demerits.

UNIT-III

Measures of Dispersion: Concept of dispersion, characteristics for an ideal measure of dispersion. Absolute and relative measures based on: range, inter quartile range, quartile deviation, coefficient of quartile deviation, Mean deviation, coefficient of mean deviation, variance, standard deviation (σ), coefficient of variation and properties of these measures, root mean square deviation and their relationship, variance of the combined series.

Moments: Moments about mean and about any point and derivation of their relationships, effect of change of origin and scale on moments, Sheppard's correction for moments (without derivation), Charlier' s checks; Pearson's β and γ coefficients.

UNIT-IV

Skewness and Kurtosis: Concept of skewness, its types, Absolute and relative (coefficient) measures of skewness with their interpretations and concept of Kurtosis, its types and coefficients of kurtosis with their interpretations.

Theory of Attributes : Symbolic notations, dichotomy of data, class frequencies, order of class frequencies, consistency of data, independence and association of attributes, Yule's coefficient of association and coefficient of colligation and their relationship.

Books recommended

S. No.	Title of Book	Name of author	Publisher
1.	Fundamental of Statistics Vol. I	Goon A.M., Gupta M.K., Dasgupta B.	World Press, Calcutta
2.	Statistics	Johnson R.	Wiley Publishers
3.	Basic Statistics International	Aggarwal B.L.	New Age
4.	Fundamentals of Mathematical Statistics	Gupta S.C.& Kapoor V.K.	Sultan Chand & Sons

(STAT-DSC-1Maj/Min-101):**DESCRIPTIVE STATISTICS-I (PRACTICAL)**

Notes:

(i) The Practical Question Paper will contain five questions. A student will be required to attempt any three questions, each of 5 marks, in three hours duration.

(ii) The practical of paper (STAT-DSC-1Maj/Min-101) will be based on the following topics:

1. To construct frequency distributions using exclusive and inclusive methods
2. Representation of data using Bar and pie diagrams
3. Representation of data using Histogram, Frequency Polygon, Frequency Curve and Ogives.
4. Graphs: stem-and-leaf-display and Box and Whisker Plot
5. To compute various measures of central tendency.
6. To compute various measures of dispersion
7. To obtain first four moments for the given grouped frequency distribution.
8. To obtain various coefficients of skewness and kurtosis.
9. To discuss the association of attributes for a 2x2 contingency table using Yule's coefficient of association and colligation.

Semester-II

Course Code: (STAT-DSC-2Maj/Min-201)
Course Name: Probability Theory

Credit: 4
L-3, P-1

Notes:

1. Students are to be engaged to 5 hours of teaching per week (3 lectures for theory and 2 lectures for practical).
2. Practical examination will be held in the respective semester.

Paper Title with code	Lecture(L)/ Credit	Practical(P)/ Credit	Marks Distribution	
Probability Theory (STAT-DSC-2 Maj/Min-201)	Lecture :3 Credit: 3	Practical Lecture:2 Credit:1	Theory	Practical
			Max. Marks: 75 Theory: 60 Internal Assessment : 15 Time: 3 Hours	Max. Marks: 25 Viva voce: 5 Record of the practical: 5 Written practical Exam: 15 Time: 3 Hours

PROBABILITY THEORY

Objective: The main objective of the course is to learn different approaches to defining probability methods. Apply probability concepts and techniques to analyze and solve problems in various fields. Also, the aim is to familiarize with random variables, probability functions, and their distributions, including discrete and continuous cases.
Gain proficiency in generating functionsand understand their properties. The focus will be on theoretical as well as practical approach
.

Learning Outcomes: On completion of this course students will be able to:

- Understand the concepts of probability, random variables and independence of random variables.
- Explain and apply various definitions and approaches to probability including classical, relative frequency, statistical, and axiomatic approaches.
- Understand and apply moments generating function, cumulant generating function, and probability generating function.
- Utilize generating functions to compute moments and cumulants.
- Analyze functions of random variables and compute joint, marginal, and conditional probability distribution functions.
- Apply different types of probability approaches to solve problems.

Note: There will be nine questions in all. Question No.1 will be compulsory covering whole of the syllabus and comprising 5 to 7 short answer type questions. Rest of the eight questions will be set from the four units uniformly i. e. two from each unit. The candidate will be required to attempt five questions in all selecting one question from each unit and the compulsory one. All the questions will carry equal marks.

UNIT-I

Concepts in Probability: Random experiment, trial, sample point, sample space, operation of events, exhaustive, equally likely, mutually exclusive and independent events; Definition of probability- classical, relative frequency, statistical and axiomatic approach.

UNIT-II

Conditional probability. Addition and multiplication laws of probability and their extension to *n* events. Boole's inequality; Baye' s theorem and its applications.

UNIT-III

Random Variable and Probability Functions: Definition of random variable, discrete and continuous random variable, probability function, probability mass function and probability density functions, distribution function and i ts properties, functions of random variables, joint, marginal and conditional probability distribution function.

UNIT-IV

Mathematical Expectation : Definition and its properties-moments, addition and multiplication theorem of expectation. Conditional expectation and conditional variance.

Generating Functions: Moments generating function, cumulant generating function, probability generating function along with their properties.Characteristic function.

Books recommended

S. No.	Title of Book	Name of author	Publisher
1.	Fundamentals of Mathematical Statistics	Gupta S.C.& Kapoor V.K.	Sultan Chand & Sons
2.	Elementary Probability	David S.	Oxford Press
3.	Introduction to the Theory of Statistics	Mood A.M, Graybill F.A and Boes D.C.	McGraw Hill.
4.	Introduction to Mathematical Statistics	Kapoor&Saxena.	S.Chand

(STAT-DSC-2 Maj/Min-201): PROBABILITY THEORY **(PRACTICAL)**

Notes:

i) The Practical Question Paper will contain five questions. A student will be required to attempt any three questions, each of 5 marks, in three hours duration.

ii) The practical of paper (STAT-DSC-2 Maj/Min-201)will be based on the following topics:

1. Problems based on classical, statistical and axiomatic definition of probability.
2. Problems based on addition and multiplication law of probability.
3. Problems based on conditional and Baye's theorem of probability.
4. Problems based on univariate random variables and distribution function.
5. Problems based on marginal and conditional probability distribution function.
6. Problems based on mathematical expectation.
7. Problem based on generating functions.

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Zoology

STUDENTS OPTING FOR SINGLE MAJOR AND SINGLE MINOR (Table1*) AND STUDENTS OPTING FOR MULTI-DISCIPLINARY/INTER-DISCIPLINARY (Table 2*) B.Sc.,B.Sc.HONOURS AND B.Sc.HONOURS WITH RESEARCH IN ZOOLOGY

Table1*						Table2*					
Sem	**Level**	**CourseCode**	**CourseName**	**Credits Th(L) + Pr(P)**	**Total Credits**	**Sem**	**Level**	**CourseCode**	**CourseName**	**Credits Th(L) + Pr(P)**	**Total Credits**
I	L-100	(ZOO-DSC-1 Maj/Min-101)	Diversity of Non-chordates	3+1=4 (x3)	24	I	L-100	(ZOO-DSC-1 Maj/Min-101)	Diversity of Non-chordates	3+1=4 (x3)	24
		(ZOO-MDC-1-101)	Health and Hygiene	2+1=3				(ZOO-MDC-1-101)	Health and Hygiene	2+1=3	
		(ZOO-VAC-1-101)	Basic Laboratory Techniques	1+1=2				(ZOO-VAC-1-101)	Basic Laboratory Techniques	1+1=2	
		(ZOO-SEC-1-101)	Vermiculture and Vermicomposting	1+2=3				(ZOO-SEC-1-101)	Vermiculture and Vermicomposting	1+2=3	
		(AEC-1-101)	Eng-1	2				(AEC-1-101)	Eng-1	2	
		(PUN-COMPL-1-101/HCP-COMPL-1-101)	Pbi-1/HCP-1	2				(PUN-COMPL-1-101/HCP-COMPL-1-101)	Pbi-1/HCP-1	2	
II	L-100	(ZOO-DSC-2 Maj/Min-201)	Diversity of Chordates	3+1=4 (x3)	24	II	L-100	(ZOO-DSC-2 Maj/Min-201)	Diversity of Chordates	3+1=4 (x3)	24
		(ZOO-MDC-2-201)	Understanding Human Body	2+1=3				(ZOO-MDC-2-201)	Understanding Human Body	2+1=3	
		(ZOO-VAC-2-201)	Vaccines and Vaccination	1+1=2				(ZOO-VAC-2-201)	Vaccines and Vaccination	1+1=2	
		(ZOO-SEC-2-201)	Apiculture	1+2=3				(ZOO-SEC-2-201)	Apiculture	1+2=3	
		(AEC-2-201)	Eng-2	2				(AEC-2-201)	Eng-2	2	
		(PUN-COMPL-2-201/HCP-COMPL-2-201)	Pbi-2/HCP-2	2				(PUN-COMPL-2-201/HCP-COMPL-2-201)	Pbi-2/HCP-2	2	
The candidate has to select Major and Minor out of 3 subjects studied in the Semesters I and II at the end of the SemesterII											
The student exiting the programme after one year,securing 48 credits, will be awarded UG Certificate provided he/she secures 4 credits from summer internship/ community work.											

***Note:Thecandidatewilloptfor eitherTemplate1or2fromSemesterIIIonwards.**

Notes:

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at the end of the second semester.
3. The student who chooses multi-disciplinary stream will continue with the three subjects and complete his/her three-year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3-year degree) shall be eligible for 4 YEAR DEGREE B.Sc. HONOURS with research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain at least 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

Semester- I (L-100)

Course Code: ZOO-DSC-1 Maj/Min-101
Course Title: Diversity of Non-Chordates

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory=75 + Practical=25)
Theory Marks= 75 {67 (Theory examination) + 08 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) + 03 (Internal assessment)}
Duration for final examination: 3 hours

Theory

Learning Outcomes:

- *The students will be able to develop an appreciation for the biodiversity of invertebrate species.*
- *The students will acquire knowledge about co-existence of different forms of living organisms ranging from unicellular to multicellular animals.*
- *The students will be able to identify different non-chordates in their surroundings.*
- *The students will be able to apply their knowledge towards identification of pathogenic non-chordates and their management.*

UNIT- 1

Protozoa: General characteristics of Phylum Protozoa
Locomotion and reproduction in *Amoeba* and *Paramecium*
Life cycle of ***Plasmodium vivax***
Porifera: General characteristics of Phylum Porifera
Canal system in *Sycon*
Types and formation of spicules in *Sycon*

UNIT- 2

Cnidaria*:* General characteristics of Phylum Cnidaria
Polymorphism in *Obelia*
Coral reefs: Types, Formation and Economic Importance
Platyhelminthes*:* General characteristics of Phylum Platyhelminthes
Life cycle of*Fasciola hepatica*
Nemathelminthes*:* General characteristics of Phylum Nemathelminthes
Life cycle of ***Ascarislumbricoides***
Parasitic adaptations in helminths

UNIT- 3

Annelida: General characteristics of Phylum Annelida
Metamerism and Reproduction in *Pheretimaposthuma*
Arthropoda: General characteristics of Phylum Arthropoda
Social organization in Termites and Honey Bee

UNIT- 4

Mollusca: General characteristics of Phylum Mollusca
Pallial complex in *Pila*
Pearl formation in Bivalves (Pearl oyster – ***Pinctadafucata***)
Echinodermata: General characteristics of Phylum Echinodermata
Water vascular system in ***Asterias***
Hemichordata: General characteristics of PhylumHemichordata
Affinities of Hemichordates with Echinoderms and Chordates

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-1 Maj/Min-101: Diversity of Non-Chordates
Practical

1. Classification up to classes with external characters, ecological notes (Habitat & Habits), and economic importance (if any) of the following animals:

- **Protozoa:***Euglena, Amoeba*, *Paramecium*, *Plasmodium*
- **Porifera:***Sycon, Hyalonema, Euplectella, Spongilla.*
- **Cnidaria:** *Obelia, Hydra, Physalia, Millepora, Aurelia, Tubipora, Alcyonium, Metridium, Fungia.*
- **Platyhelminthes:***Dugesia,Fasciola, Taenia*
- **Nemathelminthes:** *Ascaris, Ancylostoma*
- **Annelida:** *Nereis, Chaetopterus, Pheretima, Hirudinaria*
- **Arthropoda:** *Limulus*, *Palaemon*, *Cancer*, *Scolopendra*, *Julus*, *Bombyx, Periplaneta, Palamnaeus*
- **Mollusca:** *Chiton, Dentalium, Pila, Doris, Helix, Pinctada, Sepia, Octopus, Nautilus.*
- **Echinodermata:** *Asterias, Echinus, Cucumaria, Antedon*
- **Hemichordata:***Balanoglossus*

2. Study of the following permanent stained slides:

- L.S. and T.S. of*Sycon,* Spicules of *Sycon*
- Miracidium, Sporocyst, Redia, Cercaria of *Fasciola*
- Mouth parts of House fly, Butterfly, Mosquito and Cockroach.

3. Educational trip/ Project work

SUGGESTED READINGS

1. Ruppert EE and Barnes RD (2006). *Invertebrate Zoology*, VIII Edition. Holt Saunders International Edition. ISBN No.- 9780721615622.
2. Barnes RSK, Calow P, Olive PJW, Golding DW and Spicer JI (2002). *The Invertebrates: A New Synthesis*, III Edition, Blackwell Science. ISBN No.- 9781444312331.
3. Barrington EJW (1979). *Invertebrate Structure and Functions,* II Edition, Thomas Nelson & Sons Ltd. ISBN No.- 9780470265031.
4. Brusca RC, Moore W, Shuster SM (2016). *Invertebrates*, III Edition, Sinauer Associates, Inc., Publishers. ISBN No.- 9781605353753.
5. Kotpal RL (2020). *Modern Text Book of Zoology: Invertebrates,* 12th edition, Rastogi Publications. ISBN No.- 9788193887554.
6. Jordan EL andVerma PS (2022). *Invertebrate Zoology,* S. Chand and Company Ltd. ISBN No.- 9789355011954.
7. Dhami PS and Dhami, JK (2022). *A Textbook of Zoology, Vol. I Non chordates*, Pradeep Publications. ISBN No.- 9789391966003.
8. Verma PS (2010). *A Manual of Practical Zoology: Invertebrates*, S. Chand and Company Ltd.ISBN No.- 9788121908290.
9. Lal SS (2017). *Practical Zoology Invertebrate,* Rastogi Publications. ISBN No.- 9789350781616.

SEMESTER- II (L-100)

Course Code: ZOO-DSC-2 Maj/Min-201

Course Title: Diversity of Chordates

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + P=1)**
Total Marks= 100 (Theory= 75 + Practical=25)
Theory Marks= 75 {67 (Theory examination) + 08 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) + 03 (Internal assessment)}
Duration for final examination: 3 hours

Theory

Learning Outcomes:

- *Students will be able to develop an appreciation for the biodiversity of chordates.*
- *Students will acquire knowledge about origin, taxonomic classification, structure and functions of chordates.*
- *Students will understand the basic characters, advancements and adaptations of different types of vertebrates.*
- *Students will be able to apply their knowledge towards identification of different species of chordates in their surroundings.*

UNIT- 1

Introduction to Chordates
General characteristics of chordates
Differences between chordates and non-chordates
Protochordata
General characteristics of Urochordates andCephalochordates
Retrogressive metamorphosis in *Herdmania*
Agnatha
General characteristics of Cyclostomes
Affinities of Cyclostomes with protochordata and Vertebrates

UNIT- 2

Pisces
General characteristics of Chondrichthyes and Osteichthyes
Fish migration: Types and causes
Mechanism of osmoregulation in freshwater and marine fish
Amphibia
General characteristics of Class Amphibia
Parental care in amphibians
Neoteny and Paedogenesis:Types and Biological significance

UNIT- 3

Reptilia
General characteristics of Class Reptilia
Poison apparatus in snakes
Identification of poisonous and non-poisonous snakes

Aves
General characteristics of Class Aves
Archaeopteryx as a connecting link: Structural features of ***Archaeopteryx*** showing reptilian and avian characters
Flight adaptations in birds: Morphological and anatomical adaptations
Bird migration: Types and its significance

UNIT- 4

Mammalia
General characteristics of Class Mammalia
Epidermal derivatives (Hair, Nails, Claws, Hooves and Horns)
Types of Dentition (heterodont, thecodont and diphyodont)
Adaptations in mammals: Aquatic, Aerial, Arboreal, Desert and Terrestrial

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-DSC-2 Maj/Min-201: Diversity of Chordates
Practical

1. Classification up to classes with external characters, ecological notes (Habitat & Habits), and economic importance (if any) of the following animals:

- **Protochordata:** *Herdmania, Amphioxus*
- **Agnatha:** *Petromyzon, Myxine*
- **Pisces:** *Sphyrna, Pristis, Torpedo, Chimaera, Mystus, Labeo, Exocoetus, Echeneis, Anguilla(Muraena), Hippocampus, Tetraodon, Diodon, Anabas Protopterus*
- **Amphibia:** *Ichthyophis/ Ureotyphlus, Necturus, Salamandra, Hyla, Bufo*
- **Reptilia:** *Chelone, Testudo, Hemidactylus, Varanus, Chamaeleon, Draco, Python, Naja, Vipera, Hydrophis,Crocodylus*
- **Aves:** *Ardea, Milvus, Pavo, Tyto, Alcedo, Eudynamis, Casuarius*
- **Mammalia:** *Ornithorhynchus, Echidna*, *Macropus*, *Loris, Macaca, Manis*, *Hystrix*, *Funambulus*, *Canis, Herpestes*, *Capra, Pteropus*

2. Study of the following permanent stained slides:
- Larva of *Herdmania and Amphioxus*
- Scales of Fish (Cycloid, Ctenoid, Placoid)

3. Educational trip/ Project work

SUGGESTED READINGS

1. Young JZ (2004). *The Life of Vertebrates*. III Edition. Oxford University Press. ISBN No.- 9780198571735.
2. Pough H (2012). *Vertebrate life.* VIII Edition, Pearson International. ISBN No.- 9780321600790.
3. Kotpal RL (2020). *Modern Text Book of Zoology: Vertebrates, 5*th edition, Rastogi Publications. ISBN No.- 9789350780954.
4. Jordan EL andVermaPS (2021). *Chordate Zoology,* S. Chand and Company Ltd. ISBN No.- 9788121916394.
5. Dhami PS and Dhami, JK (2022). *A Textbook of Zoology, Vol. I Chordates & Evolution*, Pradeep Publications. ISBN No.- 9789391966324.
6. Verma PS (2010). *A Manual of Practical Zoology: Chordates*, S. Chand and Company Ltd.ISBN No.- 9788121908306.
7. Lal SS (2015). *Practical Zoology Vertebrate,* Rastogi Publications. ISBN No.- 9789350780169.

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